

SYDNEY GRAMMAR SCHOOL



2016

FORM V
ANNUAL EXAMINATION

Physics

Friday 2nd September 8.40AM

Working Time: 2 hours

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet, Formula Sheet and Periodic Table are provided at the back of this paper
- Write your name at the top of the Multiple Choice Answer Sheet & all pages of Part B
- Hand in your Multiple Choice Sheet and all of Part B in one bundle. (Do not staple together)

Total marks (95)

This paper has two parts: Part A and Part B.

Part A

Total marks (13)

- Attempt ALL Questions
- Allow about 20 minutes for this Part

Part B

Total marks (82)

- Attempt ALL Questions
- Allow about 1 hour 40 minutes for this Part

CHECKLIST

Each boy should have the following:

1 Question Paper

1 Multiple Choice Answer Sheet

1 – AAH	3 – PCK	5 – MRW	7 – MTK
2 – SRW	4 – DGB	6 – SRW	

EXAMINERS:

MRW/AAH/PCK/SRW/DGB

Part A**Total marks (13)****Attempt ALL Questions****Allow about 20 minutes for this Part**

Use the multiple-choice Answer Sheet.

Select the alternative A, B, C or D that best answers the question. Fill the response circle completely.

Sample

$2 + 4 =$

(A) 2

(B) 6

(C) 8

(D) 9

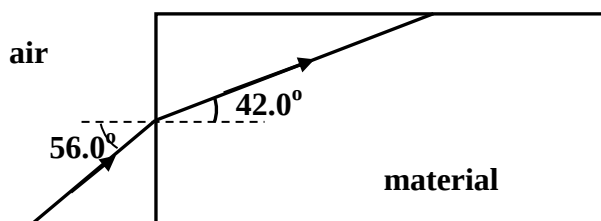
☐ A☒ B☐ C☐ D

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

☒ A☒ B☐ C☐ DIf you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.☒ A☒ B*correct*☐ C☐ D

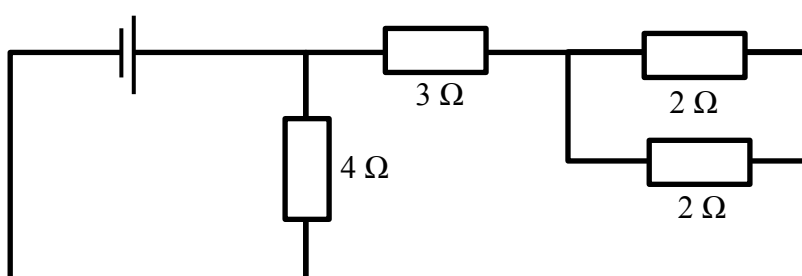
- 1 $75 \text{ km}\cdot\text{h}^{-1}$ is equivalent to:
- (A) $1.3 \text{ m}\cdot\text{s}^{-1}$
 - (B) $13 \text{ m}\cdot\text{s}^{-1}$
 - (C) $21 \text{ m}\cdot\text{s}^{-1}$
 - (D) $270 \text{ m}\cdot\text{s}^{-1}$
- 2 A train accelerates uniformly from $15 \text{ m}\cdot\text{s}^{-1}$ to $35 \text{ m}\cdot\text{s}^{-1}$ over a distance of 300 m.
The acceleration of the train is:
- (A) $0.067 \text{ m}\cdot\text{s}^{-2}$
 - (B) $0.67 \text{ m}\cdot\text{s}^{-2}$
 - (C) $1.7 \text{ m}\cdot\text{s}^{-2}$
 - (D) $3.3 \text{ m}\cdot\text{s}^{-2}$
- 3 A car of mass 1500 kg is moving at $25 \text{ m}\cdot\text{s}^{-1}$.
Its kinetic energy is:
- (A) $1.9\times 10^4 \text{ J}$
 - (B) $3.8\times 10^4 \text{ J}$
 - (C) $4.7\times 10^5 \text{ J}$
 - (D) $9.4\times 10^5 \text{ J}$
- 4 An ice-hockey puck of mass 0.16 kg is hit with an initial velocity of $18 \text{ m}\cdot\text{s}^{-1}$. It slides across a rough, horizontal surface, and takes 8.3 seconds to stop.
The magnitude of the force of friction acting on the puck is:
- (A) 0.35 N
 - (B) 2.2 N
 - (C) 3.1 N
 - (D) 14 N
- 5 When a sound wave travelled from air into a container of pure carbon dioxide gas, it was found that the speed of the wave decreased from $330 \text{ m}\cdot\text{s}^{-1}$ to $280 \text{ m}\cdot\text{s}^{-1}$.
If the frequency of the sound wave in air was 800 Hz, then it is true to say that:
- (A) the frequency of the sound wave in pure carbon dioxide was 679 Hz.
 - (B) the frequency of the sound wave in pure carbon dioxide was 750 Hz.
 - (C) the frequency of the sound wave in pure carbon dioxide was 800 Hz.
 - (D) the frequency of the sound wave in pure carbon dioxide was 943 Hz.

- 6 A ray of light travels through air and strikes a block of transparent material as shown in the following diagram.



What is the speed of the light in this material?

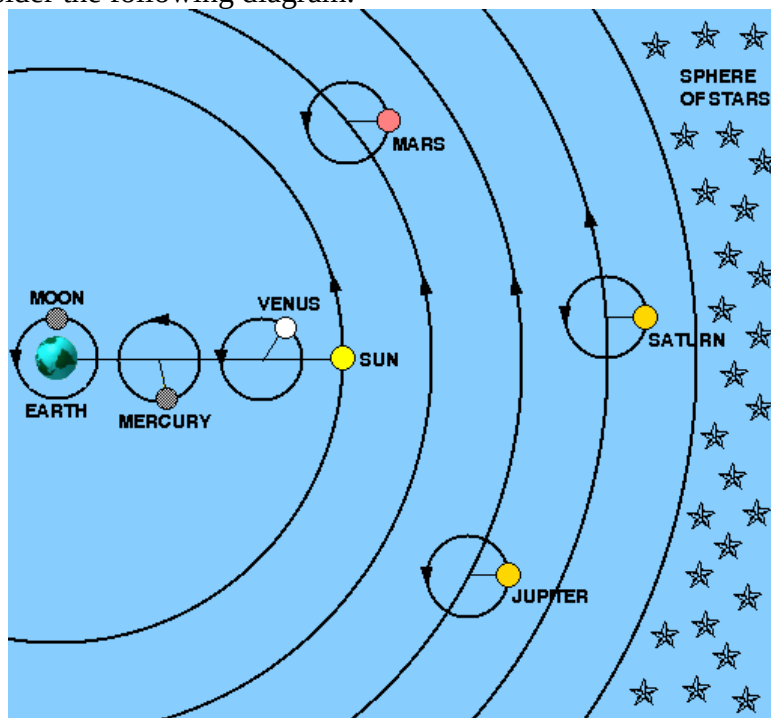
- (A) $2.28 \times 10^8 \text{ m} \cdot \text{s}^{-1}$
 (B) $2.42 \times 10^8 \text{ m} \cdot \text{s}^{-1}$
 (C) $2.51 \times 10^8 \text{ m} \cdot \text{s}^{-1}$
 (D) $2.99 \times 10^8 \text{ m} \cdot \text{s}^{-1}$
- 7 It is true to say that our Sun is a:-
- (A) white dwarf.
 (B) proto-star.
 (C) red giant.
 (D) main sequence star.
- 8 Consider the following circuit.



What fraction of the current flowing out of the battery flows through one of the 2Ω resistors?

- (A) $\frac{1}{2}$
 (B) $\frac{2}{11}$
 (C) $\frac{1}{4}$
 (D) $\frac{2}{5}$

- 9 Consider the following diagram.

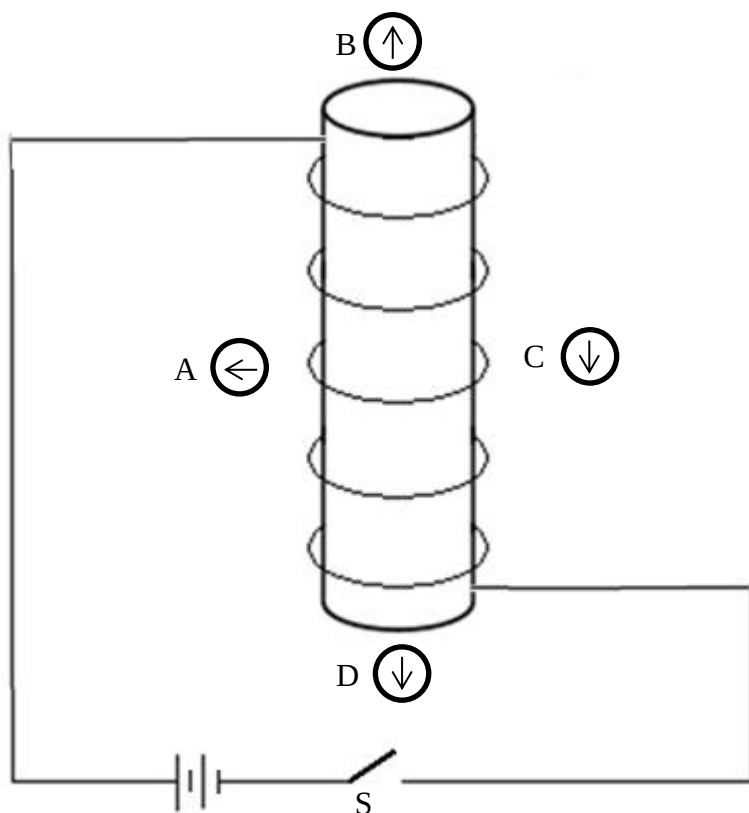


Which astronomer was responsible for the model of the Universe shown above?

- (A) Copernicus
 - (B) Kepler
 - (C) Newton
 - (D) Ptolemy
- 10 If $^{211}_{83}\text{Bi}$ decays by beta minus decay and the product of this decay then undergoes alpha decay, the final resultant nucleus is:-

- (A) $^{205}_{82}\text{Pb}$
- (B) $^{206}_{82}\text{Pb}$
- (C) $^{207}_{82}\text{Pb}$
- (D) $^{208}_{82}\text{Pb}$

- 11** The following diagram shows a soft iron bar with a coil of wire wrapped around it and connected to a switch, S and a battery. Four compasses (A, B, C and D) are placed near the coil of wire.



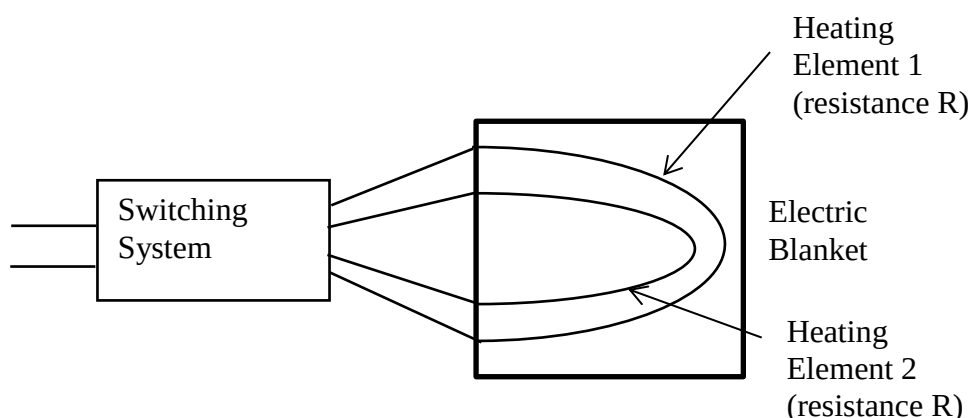
When the switch, S, is closed, which compass is correctly indicating the direction that its needle would point?

- (A) Compass A
- (B) Compass B
- (C) Compass C
- (D) Compass D

- 12 Nova radio station has a frequency of 96.9 MHz.
What is the period of the radio waves transmitted by this station?

(A) 1.03×10^{-8} s
 (B) 1.03×10^{-2} s
 (C) 3.10 s
 (D) 96.9 s

- 13 This question refers to the following diagram of a single bed electric blanket.



The electric blanket consists of two identical heating elements. In order to obtain three heat settings (high, medium and low), the blanket has a heat selector switch. The switching system is used to connect the two elements to the 240 V supply in the following combinations:-

- X: elements 1 and 2 are connected in parallel.
 Y: elements 1 and 2 are connected in series.
 Z: only element 1 is connected, element 2 is not used.

Which of the following correctly indicates the way that the two elements must be connected to provide the three heat settings?

	High	Medium	Low
(A)	X	Y	Z
(B)	X	Z	Y
(C)	Y	X	Z
(D)	Y	Z	X

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Part B**Total marks (82)****Attempt ALL Questions****Allow about 1 hour and 40 minutes for this Part**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 14 (6 marks)**Marks**

The following photograph is taken for the planet Mars over several weeks.



- (a) Name the phenomenon shown in the photograph above.

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- (b) How did Ptolemy explain this motion?

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Question 14 continued on next page.

Question 14 continued**Marks**

- (c) With the aid of a diagram, explain how Copernicus explained this motion.

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Question 15 (4 marks)**Marks**

Carbon-14 decays by beta minus decay with a half-life of 5730 years.

- (a) Write a nuclear decay equation for this reaction.

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- (b) The carbon-14 content of an ancient piece of wood was found to have only one eighth of its original carbon-14 remaining.

Determine the age (in years) of this ancient piece of wood.

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Question 16 (4 marks)**Marks**

Two of the major moons of Jupiter are Europa and Ganymede. The distance of Europa from Jupiter is 671,100 km while the distance of Ganymede from Jupiter is 1,070,400 km. The orbital period of Ganymede around Jupiter is 7.15 days.

- (a) Determine the orbital period of Europa around Jupiter.

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- (b) What was the impact of Galileo's discovery of these moons in January 1610 on the model of the Universe?

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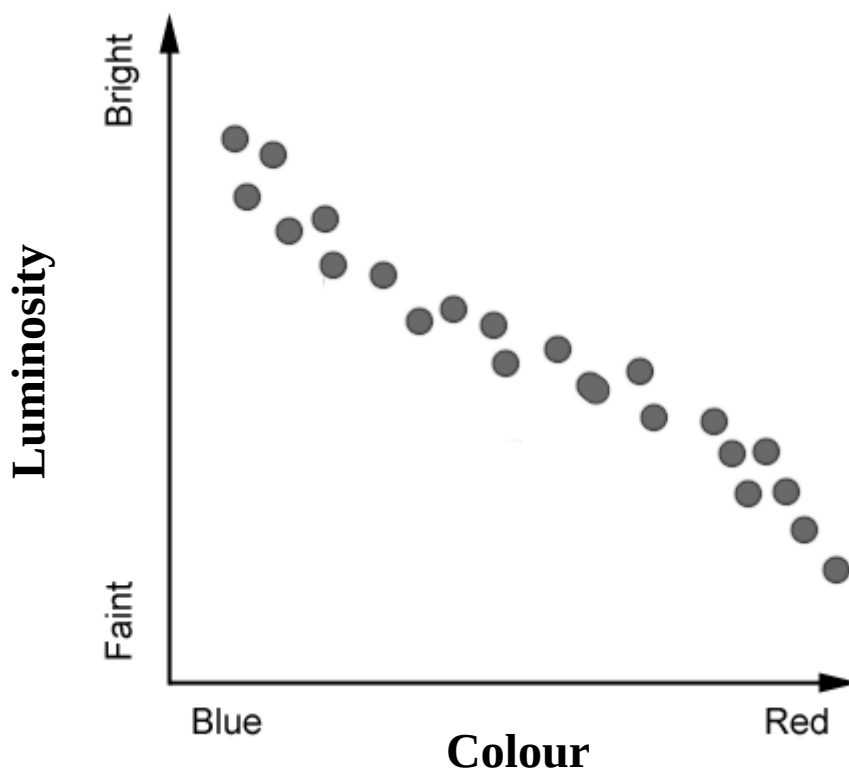
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Question 17 (3 marks)

Consider the Hertzsprung-Russell Diagram below for stars found in a globular cluster. It is known that the stars in a globular cluster are formed roughly at the same time.



- (a) Circle the hottest star shown on the diagram above.

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- (b) An astronomer looking at this diagram suggests that the stars are young. Explain the reasoning behind this suggestion.

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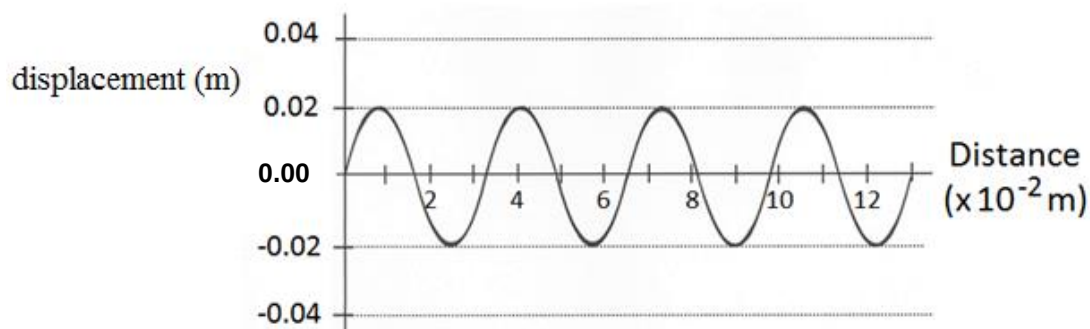
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Question 18 (4 marks)**Marks**

The diagram below shows a displacement versus distance graph for a wave travelling through a medium. The frequency of this wave is 45.0 Hz.



Use this information to find:

- (a) the wavelength of the wave.

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- (b) the amplitude of the wave.

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- (c) the period of the wave.

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- (d) the velocity of the wave.

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Question 19 (2 marks)**Marks**

The intensity of a point source of light is $40.0 \text{ W}\cdot\text{m}^{-2}$ at a distance of 8.00 m away from the source.

Calculate the intensity of the light at a distance of 12.0 m away from the same point source.

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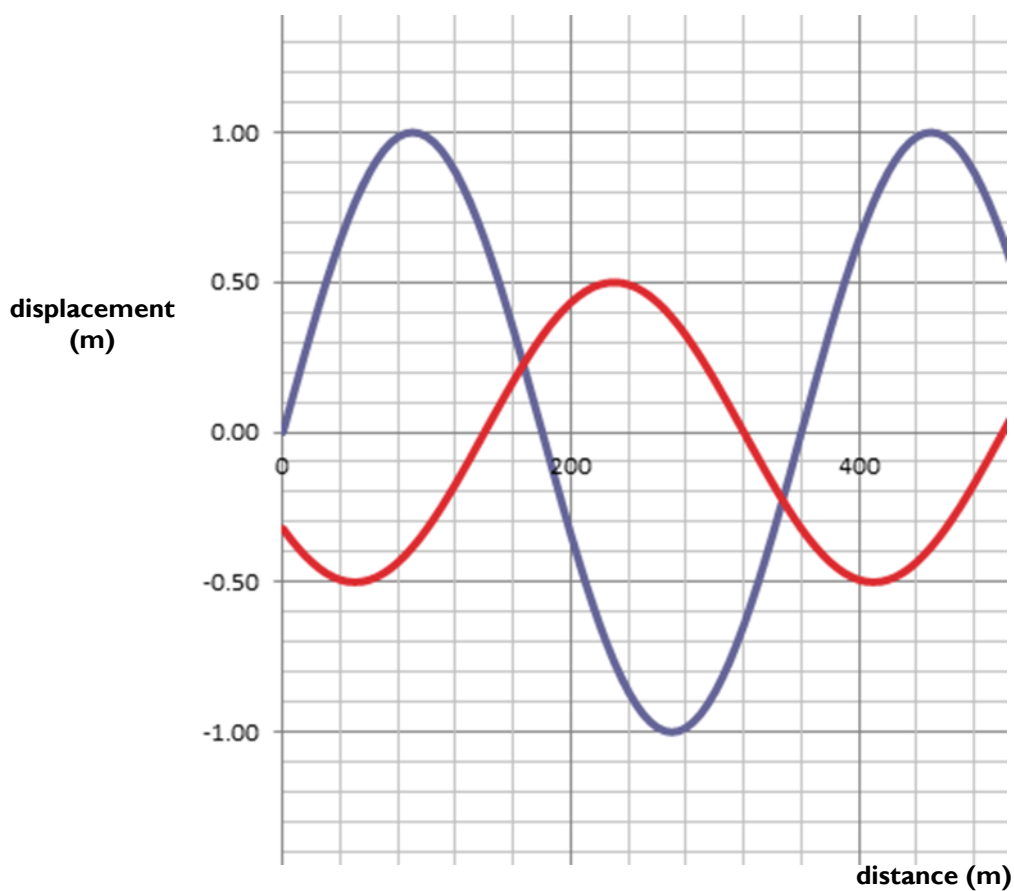
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Question 20 (2 marks)

The following graph shows a medium with two waves moving through it simultaneously.

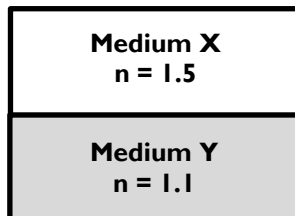


On the grid above, accurately draw the combined displacement of the medium as a result of these two waves at this instance.

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Question 21 (6 marks)

The diagram below shows a pair of transparent media, X and Y.



- (a) Using the information provided in the diagram, determine the critical angle for this pair of media.

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- (b) Determine the speed of a light ray travelling in Medium Y.

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- (c) Using a labelled diagram, describe how total internal reflection is used in a fibre optic cable.

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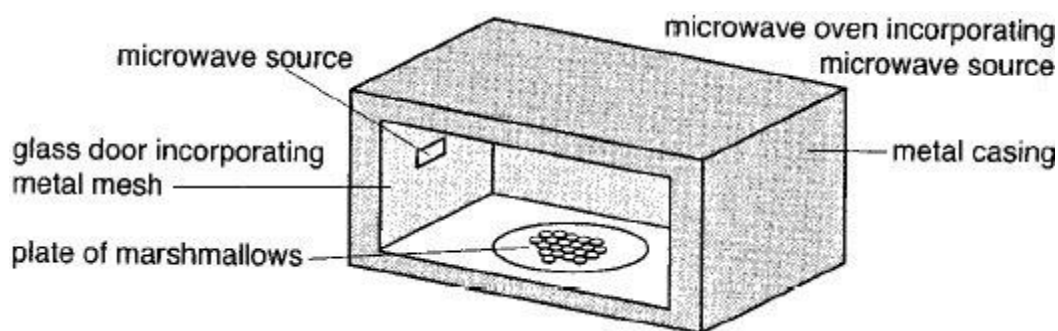
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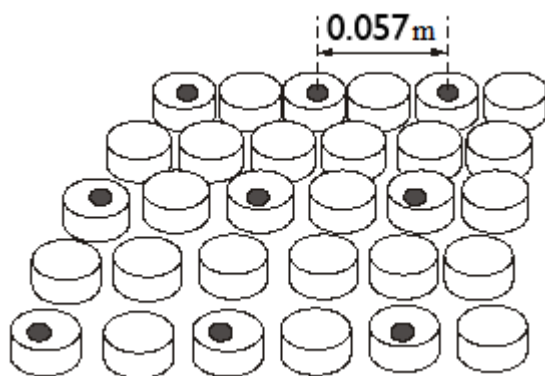
Question 22 (4 marks)**Marks**

This question is about the formation of standing waves in a microwave oven which can be used to determine the speed of light.

Marshmallow sweets can be served melted. An attempt to melt a plate of marshmallows is made in a microwave oven **from which the turntable has been removed**, so that the tray can sit still in the oven. (see Figure 1).

**Figure 1**

- (a) The marshmallows do not melt uniformly. The melting is concentrated at regularly spaced spots 0.057 m apart as shown in Figure 2.

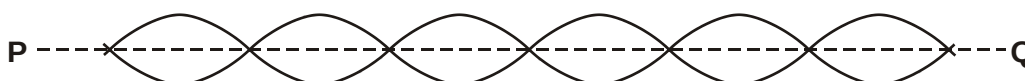
**Figure 2**

Question 22 continued on next page.

Question 22 continued**Marks**

The design of the microwave oven allows for the formation of standing waves within the cavity and the melted spots occur at the **antinodes** of a standing wave set up in the microwave oven.

Figure 3 represents the variation in amplitude of the microwaves along part of one such standing wave.

**Figure 3**

- i) Along the line **PQ**, mark with an “M”, **two** positions where a melted spot might be expected to form. **1**
- ii) On Figure 3, mark a distance equal to one wavelength. Label it λ . **1**
- (b) The label on the back of the oven states that the frequency of the microwaves is 2.45×10^9 Hz.

Calculate the speed of the **microwaves** in the microwave oven. **2**

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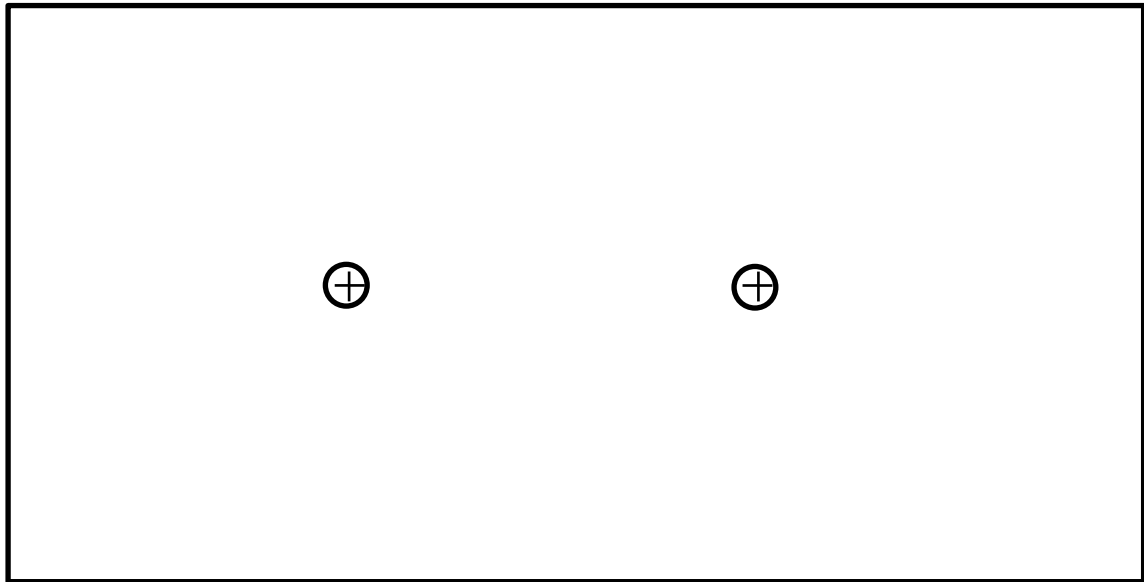
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Question 23 (4 marks)

Marks

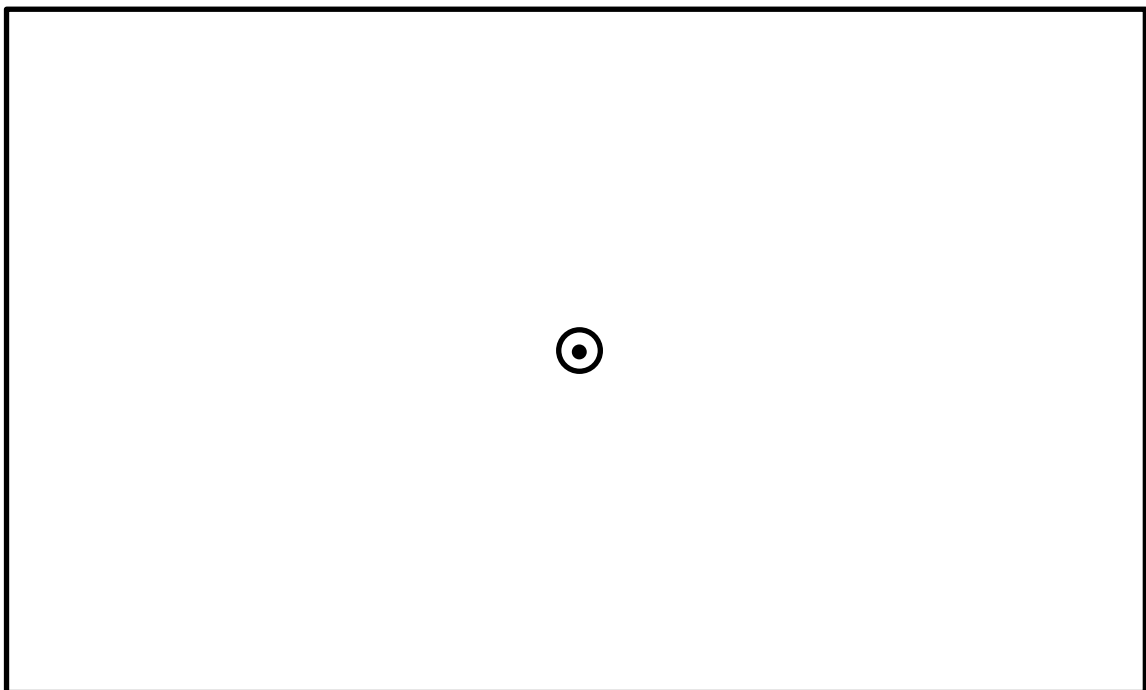
- (a) In the box provided below, draw the electric field around the two charges shown.

2



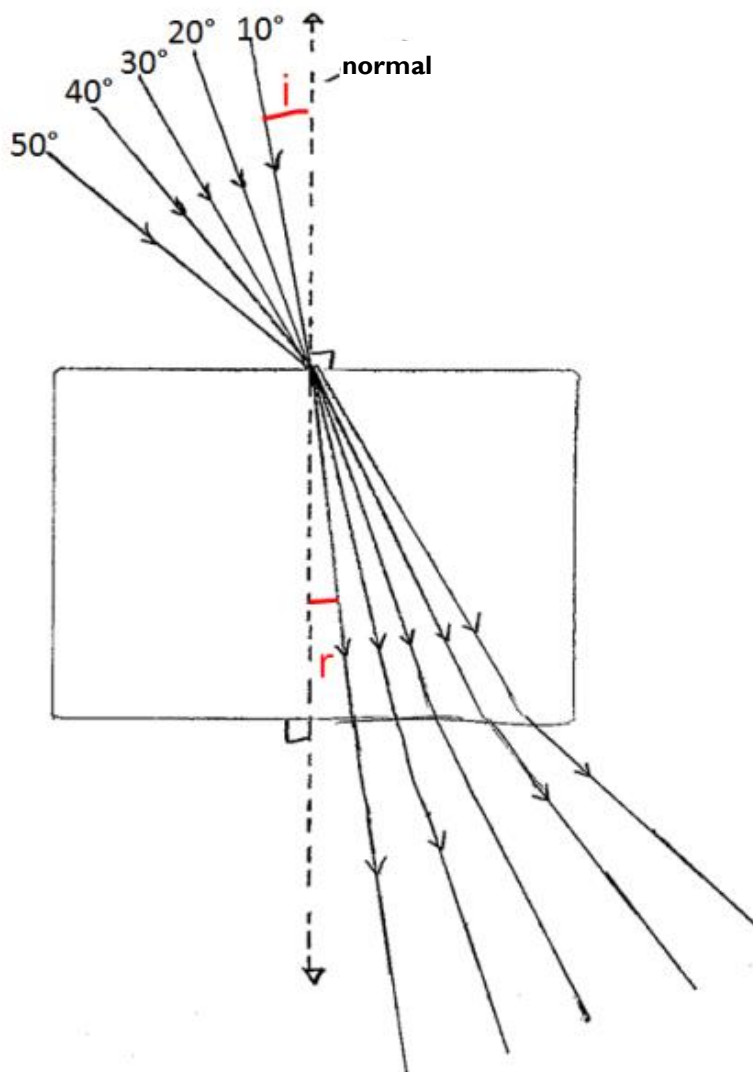
- (b) In the box provided below, draw the magnetic field around the current carrying conductor shown.

2



Question 24 (2 marks)**Marks**

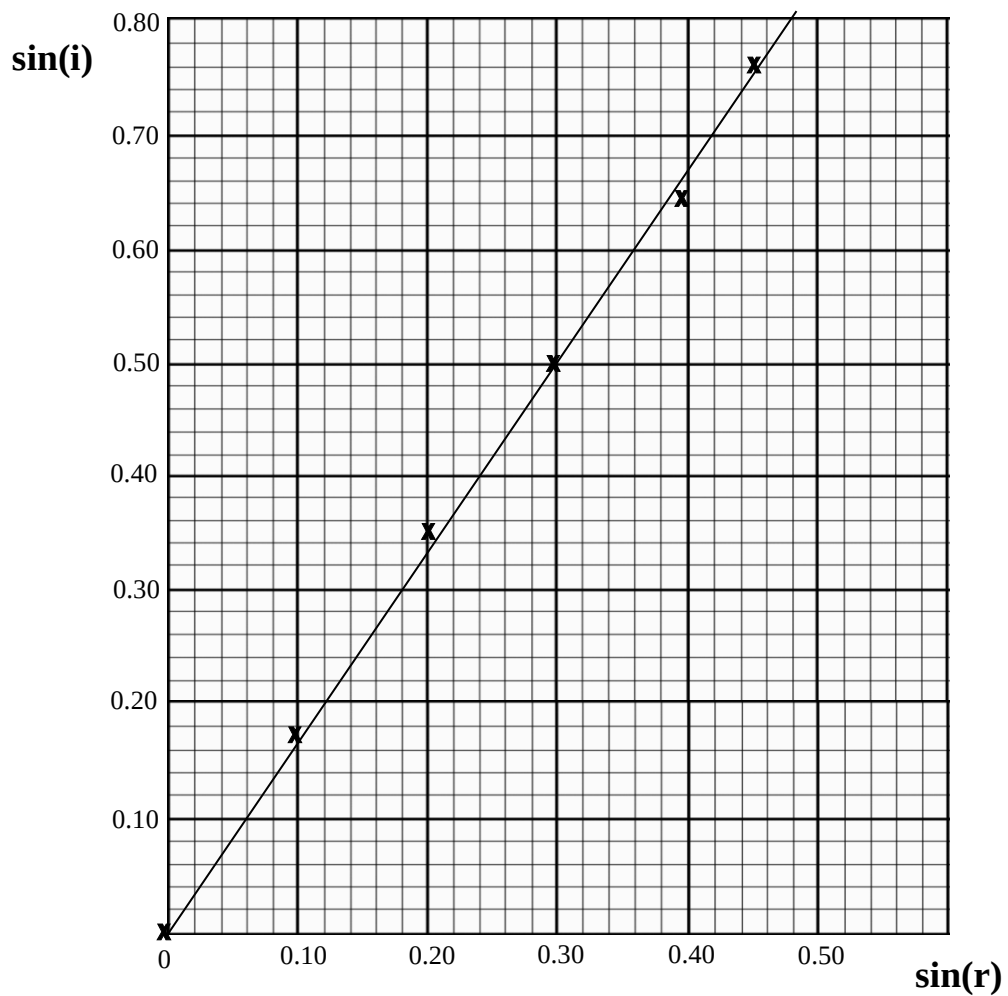
A student performed an experiment to measure the refractive index of a block of unknown transparent material. A ray of red light is shone into the block and the angles of incidence and refraction were recorded.



His results were plotted on a graph shown on the next page.

Question 24 continued on next page.

Question 24 continued



Use the graph above to determine the value of the refractive index of the block of unknown material.

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Question 25 (4 marks)**Marks**

An electric kettle when connected to a 240 V supply operates at a power of 2.40 kW. In order to boil 1.70 L of water, it must provide 0.640 MJ of energy.

- (a) Calculate the time taken to boil the water.

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- (b) What current flows through the kettle?

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- (c) A power failure occurs and the mains voltage is halved.

How long does the kettle now take to boil the water?

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Question 26 (4 marks)**Marks**

Two horizontal parallel metal plates are separated by a distance of 7.0×10^{-3} m. A potential difference of 200 V is connected across them. A small object with a mass of 2.1×10^{-12} kg is given a positive charge of 1.5×10^{-5} C. It is released from rest near the positive plate. Ignore the effects of gravity.

- (a) Calculate the magnitude of the electric field strength between the plates.

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- (b) Calculate the work done to move the $+1.5 \times 10^{-5}$ C charge through the potential difference of 200 volts.

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- (c) Calculate the speed gained by the positive charge when it reaches the negative plate.

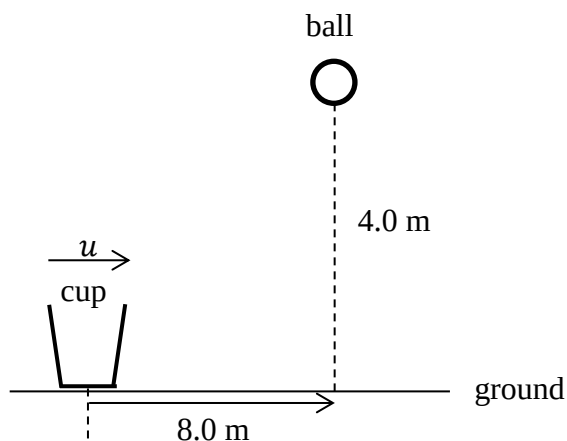
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Question 27 (5 marks)

A ball is dropped from a height of 4.0 m. At the same time, a cup is moving horizontally with an initial velocity, u , 8.0 m away as shown in the following diagram.



- (a) How long will it take for the ball to fall vertically to the ground?

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- (b) In order for the ball to land in the cup, what does the initial horizontal speed, u , of the cup have to be:-

- i) assuming the cup will slide without friction.

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- ii) assuming a frictional force acts on the cup causing it to decelerate across the surface at $1.2 \text{ m}\cdot\text{s}^{-2}$.

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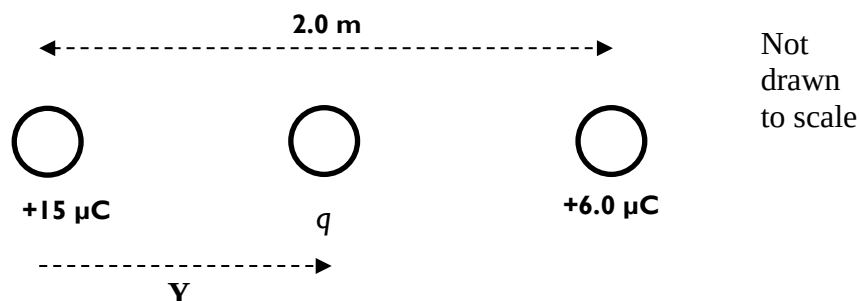
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Question 28 (4 marks)**Marks**

Two point charges, $+15\ \mu\text{C}$ and $+6.0\ \mu\text{C}$ are fixed in place and separated by $2.0\ \text{m}$.



A negative charge, q , is now placed between the two charges so that the three charges lie along a straight line as shown in the diagram above.

- (a) Calculate the position, Y , from the $+15\ \mu\text{C}$ charge where the negative charge, q , must be placed so that the resultant electrical force on it is zero.

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- (b) If at the position Y you calculated in (a) the charge q was replaced with a positive charge that is double the charge of the original q , what would be the resultant force on this new charge.

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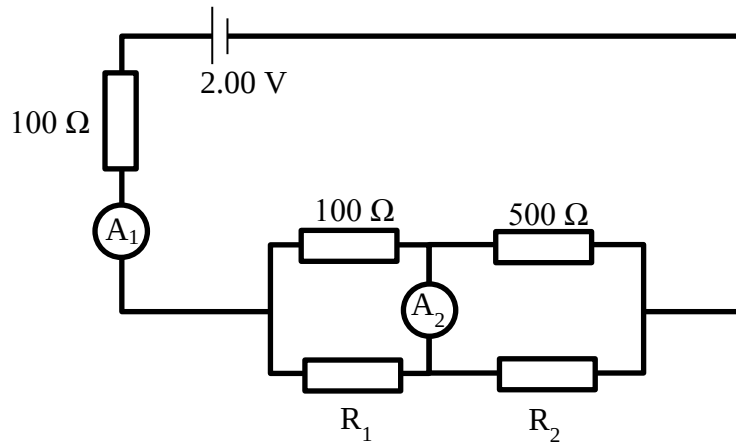
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Question 29 (3 marks)

Consider the following circuit diagram.



Ammeter, A_1 , reads 8.00 mA and ammeter, A_2 reads 0 A.

What is the value of R_1 in ohms?

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Question 30 (4 marks)**Marks**

The diagram below shows two balls A and B. Initially, ball A of mass $3m$ has a velocity of u to the right and ball B of mass m has a velocity of u to the left. The two balls A and B then collide elastically.



Calculate the values of the final velocities (a and b of balls A and B respectively) as a function of the initial speed u .

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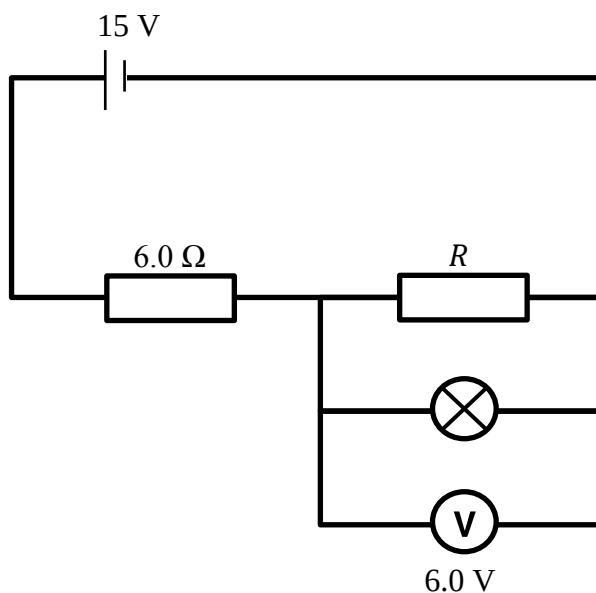
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Question 31 (4 marks)

Consider the following circuit.



A current of 0.60 A flows through the light globe.

- (a) In terms of energy, what is meant by the term ‘15 V battery’?

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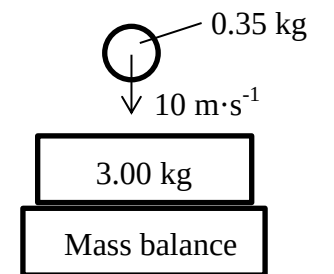
- (b) Calculate the value of the resistance labelled R .

3

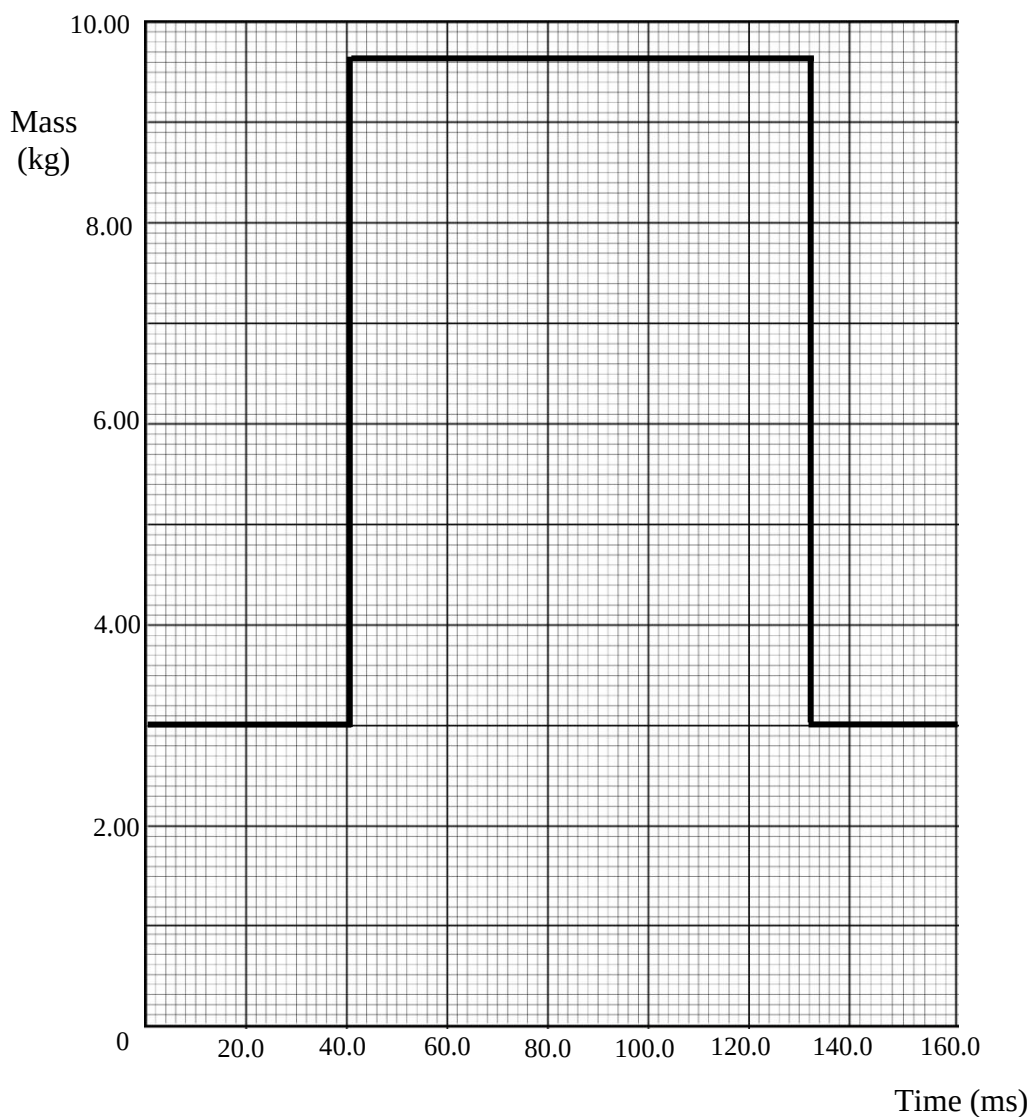
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Question 32 (5 marks)**Marks**

The diagram below shows a ball of mass 0.35 kg falling onto a 3.00 kg block which is sitting on a sensitive mass balance calibrated in kg. The ball hits the 3.00 kg block at a speed of $10\text{ m}\cdot\text{s}^{-1}$.



The graph below shows how the reading on the mass balance varies just before, during and just after the ball hits the block.



Question 32 continued on next page.

Class

Name

Question 32 continued

Marks

- (a) Calculate the magnitude of the momentum of the ball just before it hits the 3.00 kg block.

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- (b) Calculate how high the ball bounces (relative to the top of the block) after impact with the 3.00 kg block.

4

Question 33 (8 marks)**Marks**

A boy performs the following experiment to investigate friction.



The boy pulls blocks of different masses, M , along a horizontal surface with a constant force, F_0 , and measures their acceleration.

The boy assumes that the force of friction, F_F , acting on the blocks is directly proportional to the weight of the blocks, Mg , and is given by the equation:

$$F_F = \mu Mg$$

where μ is called the ‘coefficient of friction’ and can be assumed to be constant.

The results obtained by the boy are given in the table below.

Mass of block (kg)	Acceleration of block ($\text{m}\cdot\text{s}^{-2}$)		
1.00	10.5		
1.50	6.35		
2.00	4.39		
3.00	2.51		
4.00	1.42		
5.00	0.78		

The boy knows that the motion of the blocks is subject to Newton’s Second Law, and assumes that the block obeys the relationship:

$$F_0 - \mu Mg = Ma$$

which can be written as

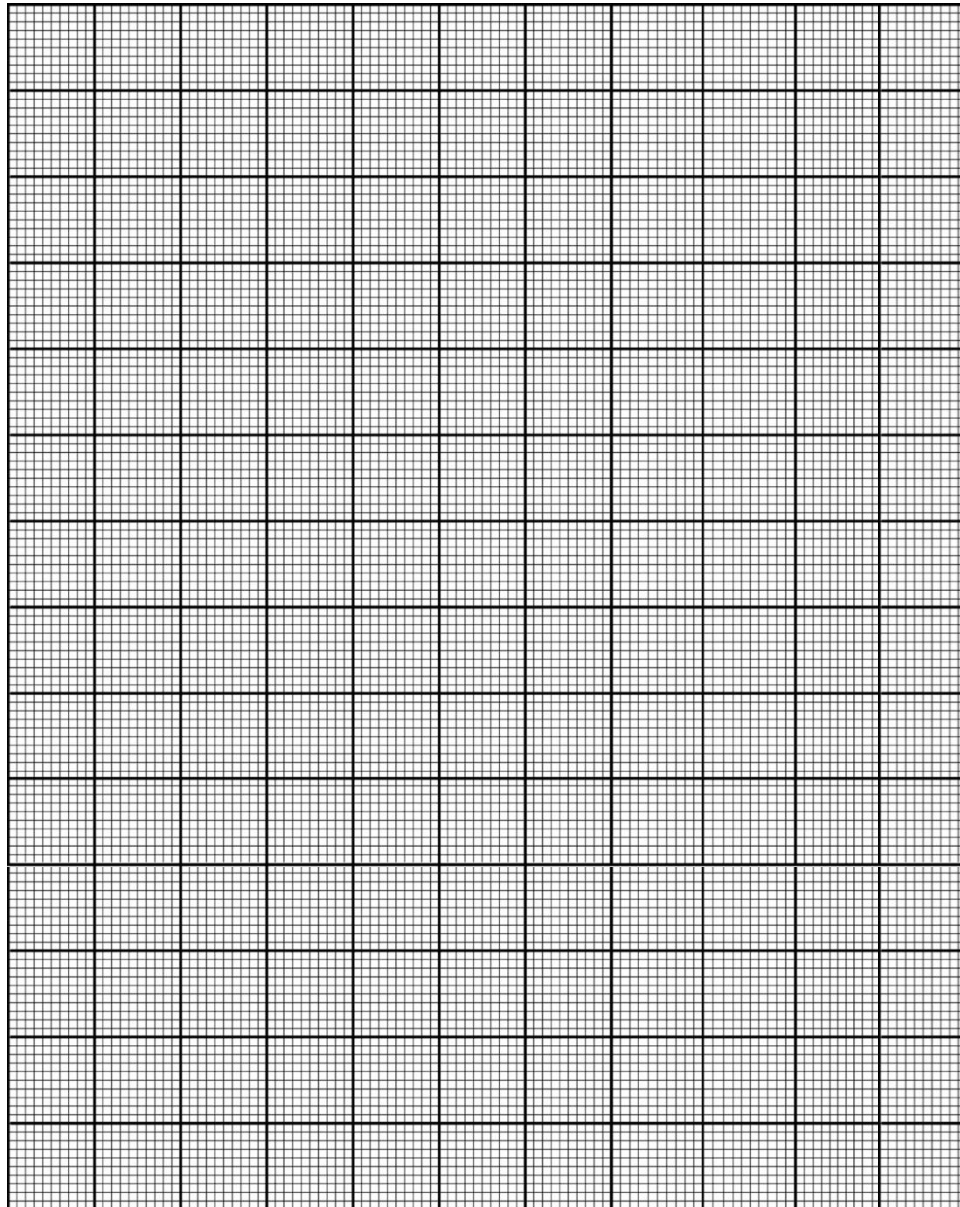
$$a = F_0 \left(\frac{1}{M} \right) - \mu g$$

Question 33 continued on next page.

Question 33 continued**Marks**

- (a) Plot a **straight line graph** to confirm the relationship between the acceleration and the mass of the blocks, given on the previous page.

(Extra columns have been provided in the table on the previous page, for you to use if you need them.)



Question 33 continued on next page.

Question 33 continued**Marks**

(b) Use your graph to determine:

i) the constant force, F_0 .

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ii) the coefficient of friction, μ .

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Physics

Data Sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	330 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Radius of Earth, R_E	$6.4 \times 10^6 \text{ m}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi}\right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ 931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$
Coulomb's constant, k	$9.0 \times 10^9 \text{ Nm}^2 \text{ C}^{-2}$

FORMULAE SHEET FORM V ONLY

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v-u}{t}$$

$$v = u + at$$

$$v^2 = u^2 + 2ar$$

$$r = ut + \frac{1}{2}at^2$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$W = Fr$$

$$p = mv$$

$$\Delta p = F_n t$$

$$F = mg$$

$$E = \frac{F}{q}$$

$$E = \frac{V}{d}$$

$$F = \frac{kQ_1Q_2}{d^2}$$

$$I = \frac{Q}{t}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$n\lambda = d \sin \theta$$

$$n\lambda = \frac{dx}{L}$$

$$E_p = -\frac{Gm_1m_2}{r}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1m_2}{d^2}$$

$$E = mc^2$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{P}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100(m_B - m_A)/5$$

$$\tau = nBIA \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$A_0 = \frac{V_{out}}{V_{in}}$$

$$c = f\lambda$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$$Z = \rho v$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

$$\begin{array}{l} \text{Surface area of a} \\ \text{sphere of radius, } R \end{array} = 4\pi R^2$$

KEY

[illegible]

Lanthanoids

57	La	138.9	Lanthanum
58	Ce	140.1	Cerium
59	Pr	140.9	Praseodymium
60	Nd	144.2	Neodymium
61	Pm		Promethium
62	Sm	150.4	Samarium
63	Eu	152.0	Europium
64	Gd	157.3	Gadolinium
65	Tb	158.9	Terbium
66	Dy	162.5	Dysprosium
67	Ho	164.9	Holmium
68	Er	167.3	Erbium
69	Tm	168.9	Thulium
70	Yb	173.1	Ytterbium
71	Lu	175.0	Lutetium

Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Elements with atomic numbers 112 and above have been reported but not fully authenticated.

Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of data. Some data may have been modified.



2016
FORM V
ANNUAL EXAMINATION

CRIB

Class

WARD

Name

Physics

Part A

ANSWER SHEET

General Instructions

- Write your class and candidate number in the space provided.
- Attempt all questions 1 – 13
- Use a blue or black pen
- Select the alternative A, B, C, or D that best answers the question.
- Fill in the response circle completely.

1. ☐ A ☐ B ☒ C ☐ D
2. ☐ A ☐ B ☒ C ☐ D
3. ☐ A ☐ B ☒ C ☐ D
4. ☒ A ☐ B ☐ C ☐ D
5. ☐ A ☐ B ☒ C ☐ D
6. ☐ A ☒ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☒ D
8. ☐ A ☐ B ☒ C ☐ D
9. ☐ A ☐ B ☐ C ☒ D
10. ☐ A ☐ B ☒ C ☐ D
11. ☐ A ☐ B ☐ C ☒ D
12. ☒ A ☐ B ☐ C ☐ D
13. ☐ A ☒ B ☐ C ☐ D

	Class
	Name

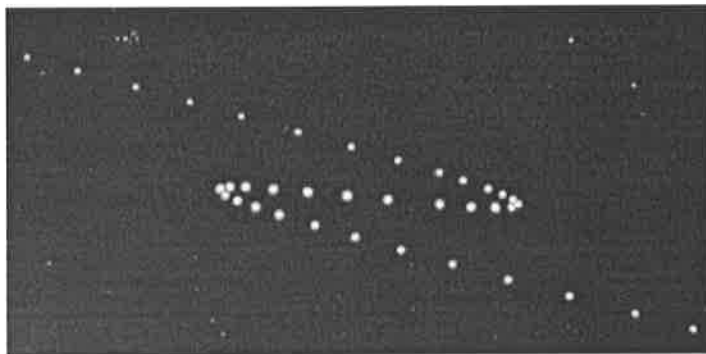
Part B**Total marks (82)****Attempt ALL Questions****Allow about 1 hour and 40 minutes for this Part**

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 14 (6 marks)**Marks**

The following photograph is taken for the planet Mars over several weeks.



- (a) Name the phenomenon shown in the photograph above.

1

Retrograde motion

- (b) How did Ptolemy explain this motion?

2

① epicycles.
① As the planet moved on its epicycle
sometimes it would be moving forward and
then backwards

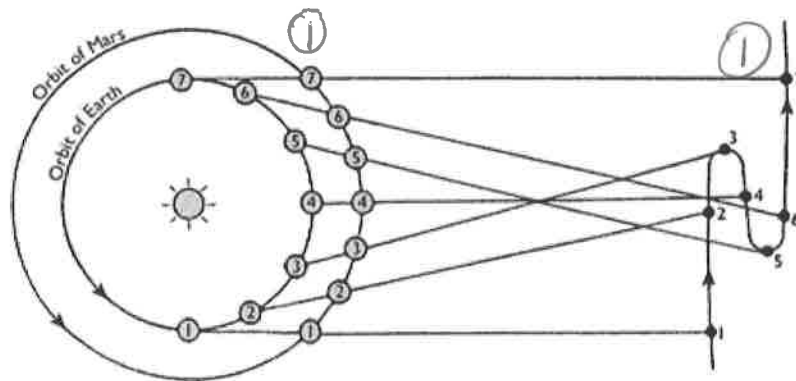
Question 14 continued on next page.

Question 14 continued

Marks

- (c) With the aid of a diagram, explain how Copernicus explained this motion.

3

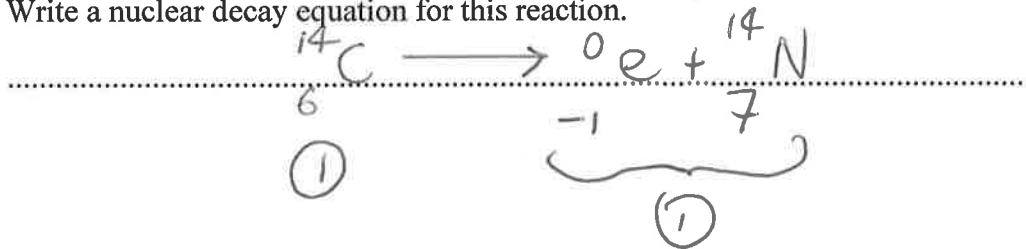


① As Earth overtakes Mars it will appear to move backwards then forwards.

Question 15 (4 marks)**Marks**

Carbon-14 decays by beta minus decay with a half-life of 5730 years.

- (a) Write a nuclear decay equation for this reaction.

2

- (b) The carbon-14 content of an ancient piece of wood was found to have only one eighth of its original carbon-14 remaining.

Determine the age (in years) of this ancient piece of wood.

2

$$\begin{array}{l} \frac{1}{8} = \left(\frac{1}{2}\right)^3 \Rightarrow 3 \text{ half lives } \textcircled{1} \\ \Rightarrow 3 \times 5730 = 17190 \text{ years } \textcircled{1} \end{array}$$

Question 16 (4 marks)**Marks**

Two of the major moons of Jupiter are Europa and Ganymede. The distance of Europa from Jupiter is 671,100 km while the distance of Ganymede from Jupiter is 1,070,400 km. The orbital period of Ganymede around Jupiter is 7.15 days.

- (a) Determine the orbital period of Europa around Jupiter.

2

$$T = T_a \sqrt{\left(\frac{r_E}{r_a}\right)^3}$$

① 3.55 days

①

- (b) What was the impact of Galileo's discovery of these moons in January 1610 on the model of the Universe?

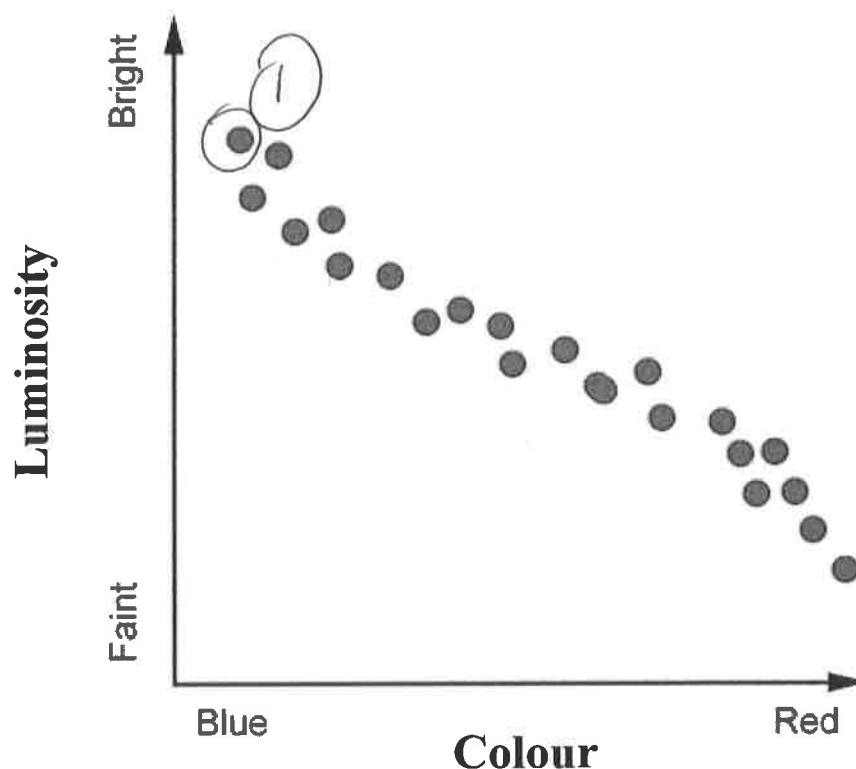
2

① As not everything is orbiting Earth

① Therefore evidence that Geocentric model is wrong.

Question 17 (3 marks)**Marks**

Consider the Hertzsprung-Russell Diagram below for stars found in a globular cluster. It is known that the stars in a globular cluster are formed roughly at the same time.



- (a) Circle the hottest star shown on the diagram above.

1

- (b) An astronomer looking at this diagram suggests that the stars are young. Explain the reasoning behind this suggestion.

2

① Only main sequence stars.

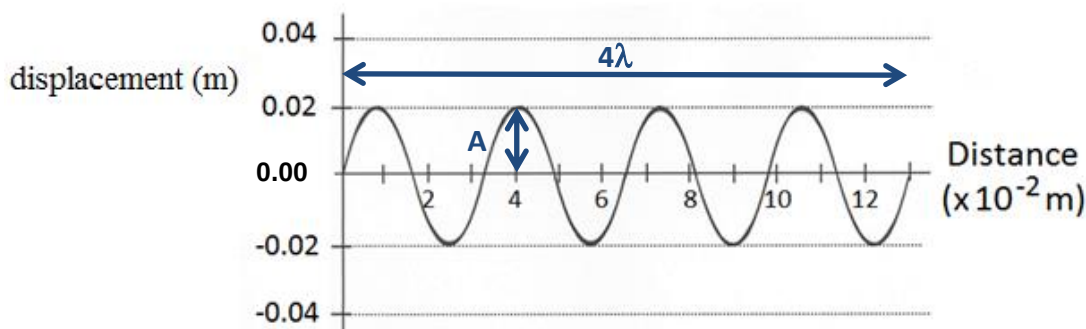
① no red giants, white dwarfs etc
that are evidence of older stars

- DGB -

Question 18 (4 marks)

Marks

The diagram below shows a displacement versus distance graph for a wave travelling through a medium. The frequency of this wave is 45.0 Hz.



Use this information to find:

- (a) the wavelength of the wave.

1

Criteria	Marks
• Correct value determined (range: ± 0.25 cm accepted).	1 (91%)

Sample answer: From the graph, $4\lambda = 13 \times 10^{-2} \text{ m}$
 so $\lambda = 13 \times 10^{-2} / 4 = \underline{3.25 \times 10^{-2} \text{ m}}$ ($= 0.00325 \text{ m} = 3.25 \text{ cm}$)

- (b) the amplitude of the wave.

1

Criteria	Marks
• Correct value determined.	1 (97%)

Sample answer: $A = \underline{0.02 \text{ m}}$

- (c) the period of the wave.

1

Criteria	Marks
• Correct value determined.	1 (99%)

Sample answer: $T = \frac{1}{f} = \frac{1}{45.0} = \underline{2.22 \times 10^{-2} \text{ s}}$

- (d) the velocity of the wave.

1

Criteria	Marks
• Correct value determined [carry-over errors from (a) allowed].	1 (94%)

Sample answer: $v = f \lambda = 45.0 \times 3.25 \times 10^{-2} = \underline{1.46 \text{ m.s}^{-1}}$

Question 19 (2 marks)**Marks**

The intensity of a point source of light is $40.0 \text{ W}\cdot\text{m}^{-2}$ at a distance of 8.00 m away from the source.

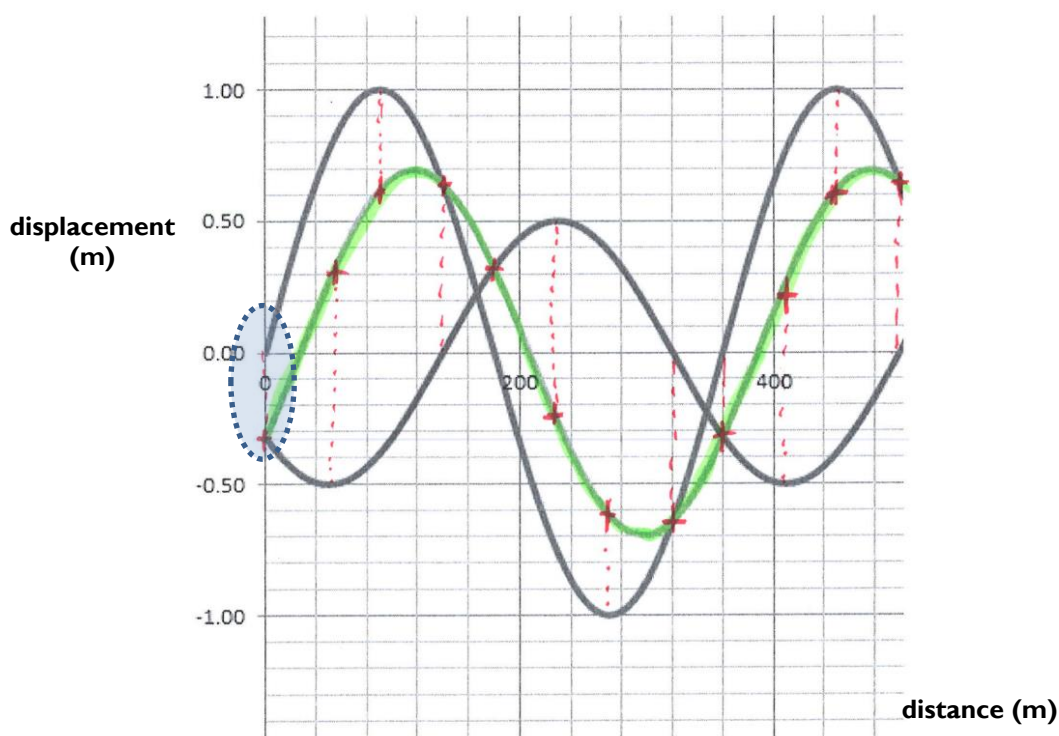
Calculate the intensity of the light at a distance of 12.0 m away from the same point source. 2

Criteria	Marks
• Appropriate equation used. • Correct substitution and result.	2 (90%)
• Appropriate equation used but calculation error.	1 (7%)

Sample answer: $I_1 d_1^2 = I_2 d_2^2 \Rightarrow I_2 = I_1 \frac{d_1^2}{d_2^2} = 40.0 \times \frac{8.0^2}{12.0^2} = \underline{17.8 \text{ W}\cdot\text{m}^{-2}}$

Question 20 (2 marks)

The following graph shows a medium with two waves moving through it simultaneously. 2



Criteria	Marks
• Correct shape. • Reasonable accuracy.	2 (80%)
• Any mistakes: -1 mark (especially the first point on the left ).	1 (19%)

On the grid above, accurately draw the combined displacement of the medium as a result of these two waves at this instance.

Question 21 (6 marks)**Marks**

The diagram below shows a pair of transparent media, X and Y.

Medium X $n = 1.5$
Medium Y $n = 1.1$

- (a) Using the information provided in the diagram, determine the critical angle for this pair of media. 2

Criteria	Marks
- Appropriate equation used. - Correct substitution ($r = 90^\circ$ when $i = i_c$) and result.	2 (94%)
Appropriate equation but calculation error.	1 (4%)

Sample answer: $n_x \sin i_c = n_y \sin 90^\circ \Rightarrow i_c = \sin^{-1} \left(\frac{n_y}{n_x} \sin 90^\circ \right) = \sin^{-1} \left(\frac{1.1}{1.5} \times 1 \right) = \underline{47.17^\circ}$

- (b) Determine the speed of a light ray travelling in Medium Y. 1

Criteria	Marks
Correct value determined.	1 (88%)

Sample answer: $n_y = \frac{c}{v_y} \Rightarrow v_y = \frac{c}{n_y} = \frac{3.0 \times 10^8}{1.1} = \times 3.25 \times 10^{-2} = \underline{2.72 \times 10^8 \text{ m.s}^{-1}}$

- (c) Using a labelled diagram, describe how total internal reflection is used in a fibre optic cable. 3

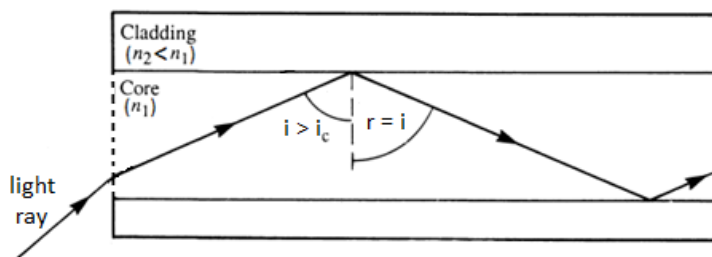
Criteria	Marks
- Appropriate diagram (with labels), showing important angles and the different parts of the optic fibre (cladding + core, and respective indices of refraction). AND - Clear and thorough explanation of how Total Internal Reflection (TIR) is used in optic fibres, especially in terms of angle of incidence ($i > i_c$) and of relative indices of refraction between the two media ($n_{\text{cladding}} < n_{\text{core}}$).	3 (37%)
One of the above bullet points missing or poorly described.	2 (27%)
Any appropriate comment (or element in the diagram) showing any basic understanding of how Total Internal Reflection is used in optic fibres.	1 (26%)

Marker's comments:

- Many boys just stated that optical fibres could transfer information by using Total Internal Reflection, usually with a poorly drawn diagram (missing cladding or light rays or angles or respective indices of refraction).
- Angles are ALWAYS measured from the normal to the surface.

Sample answer:

An optical fibre is comprised of a light-carrying core (higher refractive index) surrounded by a cladding (lower refractive index).



This construction traps the light in the core by the principle of Total Internal Reflection.

Since the fibre core has a slightly higher refractive index, the light in the core is totally reflected at the boundary of the cladding for all light that strikes at greater angles than the critical angle for this pair of materials (which should be as small as possible).

As a consequence, very little light is absorbed by the glass: the light getting in at one end undergoes repeated total internal reflection (even when the fibre is bent) and emerges at the other end.

Question 22 (4 marks)

Marks

This question is about the formation of standing waves in a microwave oven which can be used to determine the speed of light.

Marshmallow sweets can be served melted. An attempt to melt a plate of marshmallows is made in a microwave oven **from which the turntable has been removed**, so that the tray can sit still in the oven. (see Figure 1).

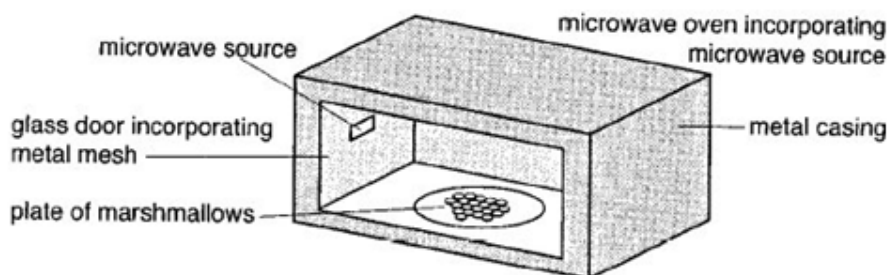


Figure 1

- (a) The marshmallows do not melt uniformly. The melting is concentrated at regularly spaced spots 0.057 m apart as shown in Figure 2.

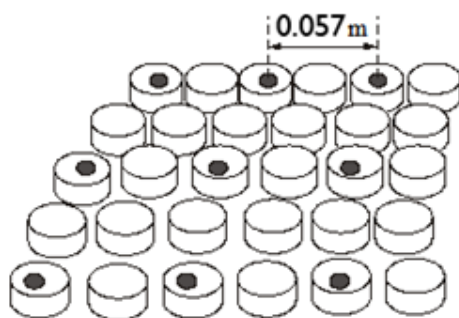


Figure 2

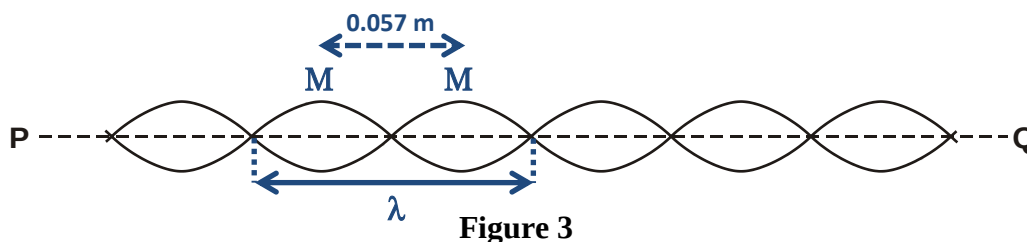
Question 22 continued on next page.

Question 22 continued

Marks

The design of the microwave oven allows for the formation of standing waves within the cavity and the melted spots occur at the **antinodes** of a standing wave set up in the microwave oven.

Figure 3 represents the variation in amplitude of the microwaves along part of one such standing wave.



- i) Along the line **PQ**, mark with an “M”, **two** positions where a melted spot might be expected to form.

1

Criteria	Marks
• Correct positions marked (M).	1 (54%)

Sample answer: see diagram above.

- ii) On Figure 3, mark a distance equal to one wavelength. Label it λ .

1

Criteria	Marks
• Correct distance marked (λ).	1 (54%)

Sample answer: see diagram above.

- (d) The label on the back of the oven states that the frequency of the microwaves is 2.45×10^9 Hz.

Calculate the speed of the **microwaves** in the microwave oven.

2

Criteria	Marks
• Appropriate equation used.	2
• Correct substitution ($\lambda = 2 \times 0.057 = 0.114$ m) and result.	(74%)
• Appropriate equation but calculation error (mainly, $\lambda = 0.057$ m).	1 (23%)

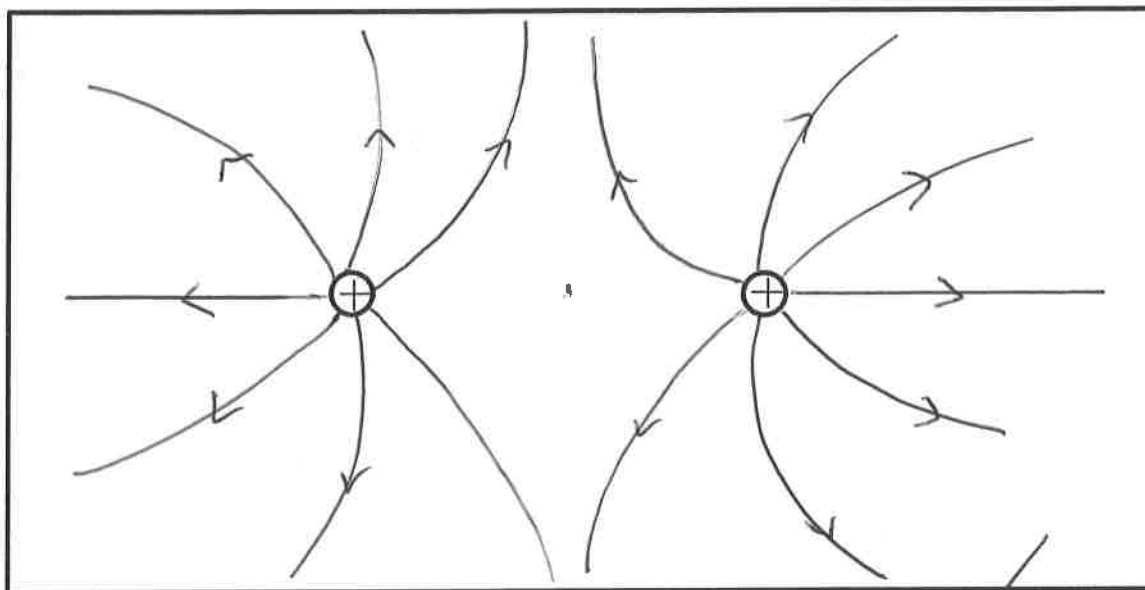
Sample answer: $v = f \lambda$ with $\lambda = 2 \times 0.057 \text{ m} = 0.114 \text{ m}$ (see diagram above)

$$\Rightarrow v = 2.45 \times 10^9 \times 0.114 = \underline{2.79 \times 10^8 \text{ m.s}^{-1}}$$

Question 23 (4 marks)**Marks**

- (a) In the box provided below, draw the electric field around the two charges shown.

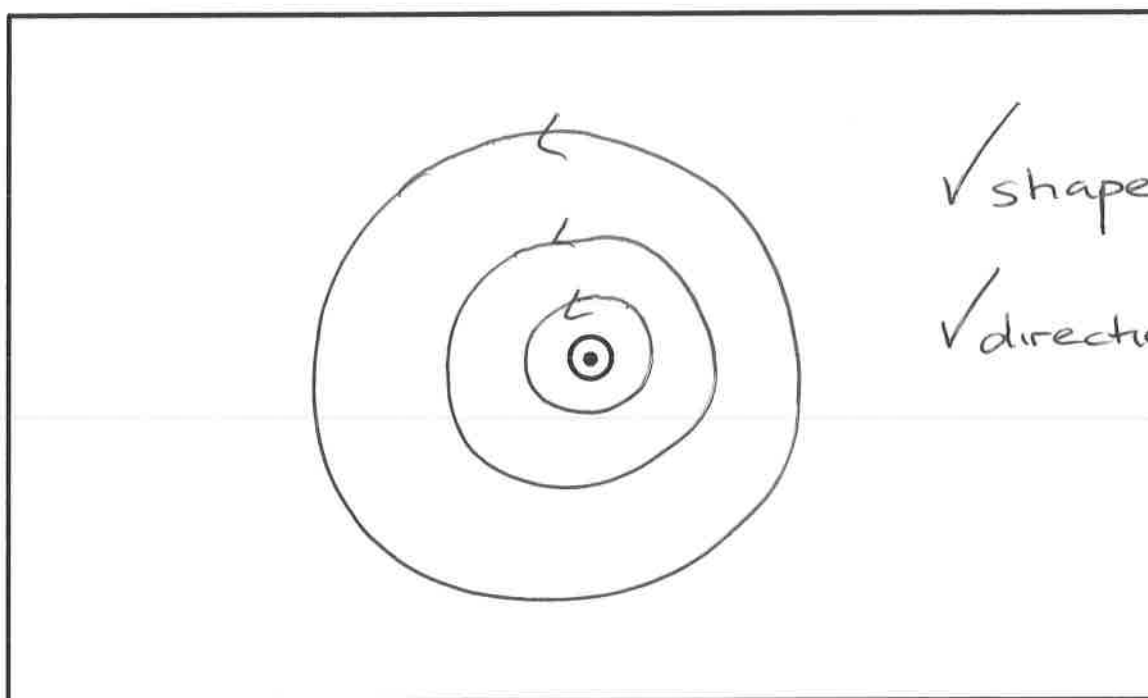
2



✓ shape
✓ direction

- (b) In the box provided below, draw the magnetic field around the current carrying conductor shown.

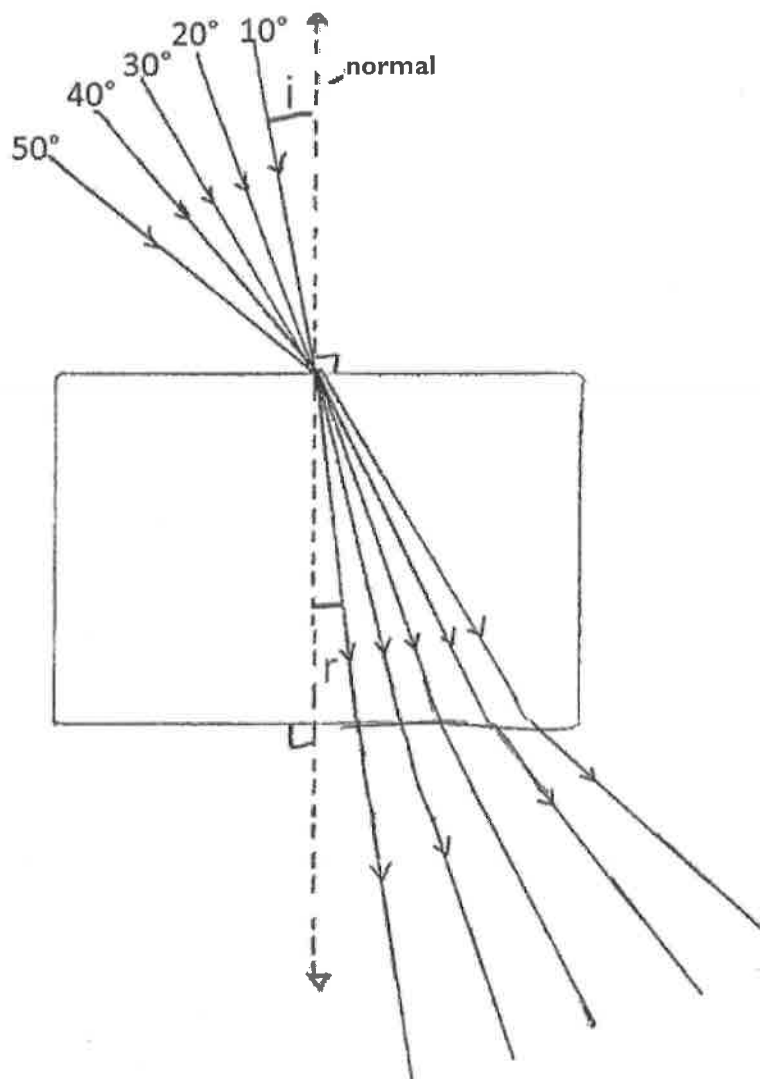
2



✓ shape
✓ direction

Question 24 (2 marks)**Marks**

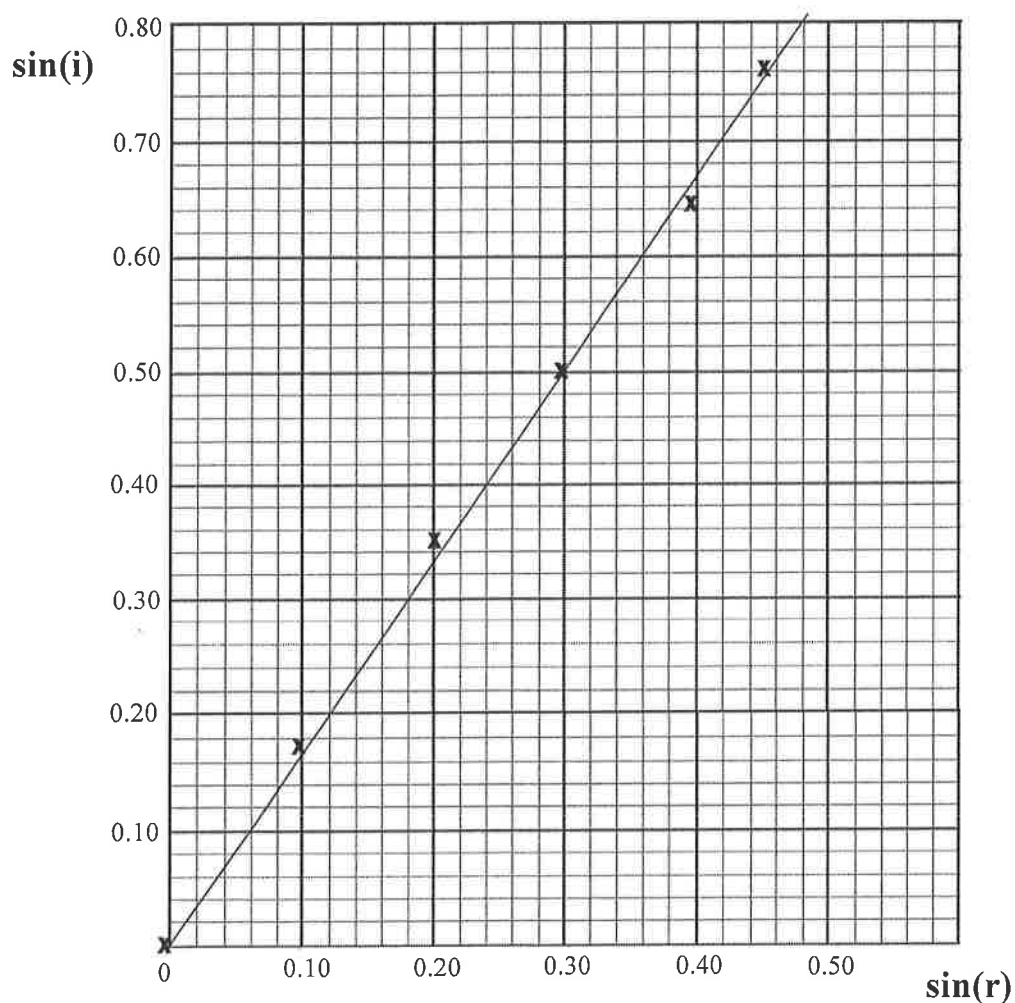
A student performed an experiment to measure the refractive index of a block of unknown transparent material. A ray of red light is shone into the block and the angles of incidence and refraction were recorded.



His results were plotted on a graph shown on the next page.

Question 24 continued on next page.

Question 24 continued



Use the graph above to determine the value of the refractive index of the block of unknown material.

2

$$n = \sin i / \sin r$$

$$= 0.50 / 0.30$$

$$= 1.7$$

-1 mk for each mistake made.

if $\frac{\sin 0.50}{\sin 0.30}$

then $\sqrt{x}$ 1mk for recognition of ratio but incorrect substitution

Question 25 (4 marks)**Marks**

An electric kettle when connected to a 240 V supply operates at a power of 2.40 kW. In order to boil 1.70 L of water, it must provide 0.640 MJ of energy.

- (a) Calculate the time taken to boil the water.

2

$$EE = VIt$$

$$0.640 \times 10^6 = 240 \times \frac{2400}{240} \times t$$

$$t = 267s$$

$$\text{or } E = P \times t$$

$$0.640 \times 10^6 = 2.4 \times 10^3 t$$

✓ correct substitution
✓ correct answer based on substitution.

- (b) What current flows through the kettle?

1

$$P = VI$$

$$2.40 \times 10^3 = 240I \quad \therefore I = 10A. \quad \text{✓ ①mk.}$$

[correct substitution into a correct formula]

Answer only.

- (c) A power failure occurs and the mains voltage is halved.

How long does the kettle now take to boil the water?

1

$$EE = P \times t \quad (4 \times \text{longer})$$

$$0.640 \times 10^6 = \frac{120^2}{24} \times t \quad t = 1066.7s \quad \text{✓ ①mk.}$$

for answer
4 x Ans(a)

$$[\text{or } 0.640 \times 10^6 = \frac{240}{2} \times 5 \times t]$$

Note: from $V = IR$

$$240 = 10 \times R$$

$$R = 24\Omega$$

if V halved,
then since R
= 24Ω is constant,
 $I = 5A$.

Most candidates
assumed I was
still equal to $10A$.

$$\therefore t = 533.33s$$

X 0mks

Question 26 (4 marks)**Marks**

Two horizontal parallel metal plates are separated by a distance of 7.0×10^{-3} m. A potential difference of 200 V is connected across them. A small object with a mass of 2.1×10^{-12} kg is given a positive charge of 1.5×10^{-5} C. It is released from rest near the positive plate. Ignore the effects of gravity.

- (a) Calculate the magnitude of the electric field strength between the plates.

1

$$E = V/d = 200 / 7 \times 10^{-3} = 2.86 \times 10^4 \text{ NC}^{-1}$$

✓ 1mk correct substitution

- (b) Calculate the work done to move the
- $+1.5 \times 10^{-5}$
- C charge through the potential difference of 200 volts.

1

$$W = qV = 1.5 \times 10^{-5} \times 200 = 3 \times 10^{-3} \text{ J}$$

Or $W = qEd = 1.5 \times 10^{-5} \times \text{Ans(a)} \times 7.0 \times 10^{-3}$

- (c) Calculate the speed gained by the positive charge when it reaches the negative plate.

CE = carry through error 2

Method 1: $W = \Delta KE$

$$\text{Ans(b)} = \frac{1}{2} \times 2.1 \times 10^{-12} v^2$$

$$v = 5.3 \times 10^4 \text{ ms}^{-1}$$

-1 mark for each mistake

Method 2:

$$F = ma = qE$$

$$2.1 \times 10^{-12} a = 1.5 \times 10^{-5} \times \text{Ans(a)}$$

$$a = 2.04 \times 10^{11} \text{ ms}^{-2}$$

✓ 1mk. for correct value.

$$v^2 = u^2 + 2as$$

$$v^2 = 0^2 + 2 \times 2.04 \times 10^{11} \times 7 \times 10^{-3}$$

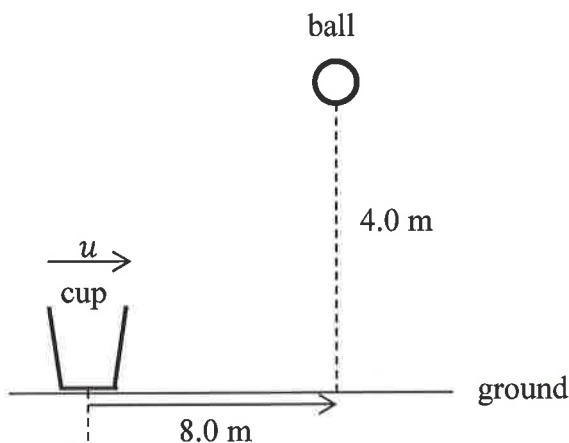
$$v = 5.35 \times 10^4 \text{ ms}^{-1}$$

✓ 1mk for correct substitution

Note: if wrong answer with no working = 0 mks.

Question 27 (5 marks)**Marks**

A ball is dropped from a height of 4.0 m. At the same time, a cup is moving horizontally with an initial velocity, u , 8.0 m away as shown in the following diagram.



- (a) How long will it take for the ball to fall vertically to the ground?

2

$$\begin{aligned}
 u &= 0 & r &= \frac{1}{2}at^2 \\
 a &= -9.8 \text{ m/s}^2 & t &= \sqrt{\frac{2r}{a}} \\
 r &= -4 \text{ m} & &= \sqrt{\frac{2 \times (-4)}{-9.8}} = 0.90 \text{ s} \quad \text{(2 sig fig)}
 \end{aligned}$$

① solve for time and use $g = 9.8 \text{ m/s}^2$
 ① correct answer

- (b) In order for the ball to land in the cup, what does the initial horizontal speed, u , of the cup have to be:-

- i) assuming the cup will slide without friction.

1

$$r = 8 \text{ m}, t = 0.90 \quad r = ut, \quad u = \frac{r}{t} = \frac{8}{0.90} = 8.9 \text{ (2 sig fig)}$$

① mark for correct answer

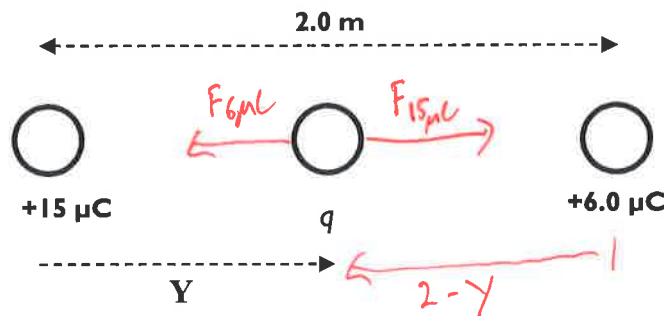
- ii) assuming a frictional force acts on the cup causing it to decelerate across the surface at 1.2 m/s^2 .

2

$$\begin{aligned}
 r &= ut + \frac{1}{2}at^2 & a &= -1.2 \text{ m/s}^2 \\
 u &= \frac{r - \frac{1}{2}at^2}{t} & & \\
 &= \frac{r}{t} - \frac{1}{2}at = \frac{8.0}{0.90} - \frac{(-1.2)}{2} (0.90) & \text{① mark for correctly using } a = -1.2 \text{ m/s}^2 \text{ in correct equation} \\
 &= \frac{8.0}{0.90} + \frac{1.2}{2} (0.90) = 9.4 \text{ m/s (2 sig fig)} \\
 & & \text{① mark for correct answer}
 \end{aligned}$$

Question 28 (4 marks)**Marks**

Two point charges, $+15 \mu\text{C}$ and $+6.0 \mu\text{C}$ are fixed in place and separated by 2.0 m .



Not
drawn
to scale

many answers do not
recognise that there are
2 forces on q from each charge
and the net force
is the sum of these
forces.

A negative charge, q , is now placed between the two charges so that the three charges lie along a straight line as shown in the diagram above.

- (a) Calculate the position, Y , from the $+15 \mu\text{C}$ charge where the negative charge, q , must be placed so that the resultant electrical force on it is zero.

$$\Sigma F = 0 = \frac{k(15 \times 10^{-6})(q)}{Y^2} - \frac{k(6 \times 10^{-6})(q)}{(2-Y)^2}$$

① Equating the forces from $15 \mu\text{C}$ and $6 \mu\text{C}$

$$\frac{k(15 \times 10^{-6})(q)}{Y^2} = \frac{k(6 \times 10^{-6})(q)}{(2-Y)^2}$$

equating, cancel out k and q

$$\frac{15}{Y^2} = \frac{6}{(2-Y)^2}$$

multiply by denominator

$$15(2-Y)^2 = 6Y^2$$

$$15Y^2 - 60Y + 60 = 6Y^2$$

$$9Y^2 - 60Y + 60 = 0 \quad \text{or} \quad 3Y^2 - 20Y + 20 = 0$$

quadratic

$$Y = \frac{20 \pm \sqrt{20^2 - 4(3)(20)}}{2 \times 3}$$

$$Y = 1.2 \text{ (2 sig fig)} \text{ or } 5.4 \text{ (2 sig fig)}$$

only $Y = 1.2$ is a valid solution - ($0 < Y < 2$)

① getting a solvable quadratic

① Giving only correct valid solution

(38% of form 3/3, 37% 6/3)

- (b) If at the position Y you calculated in (a) the charge q was replaced with a positive charge that is double the charge of the original q , what would be the resultant force on this new charge.

$$\Sigma F \text{ is Zero}$$

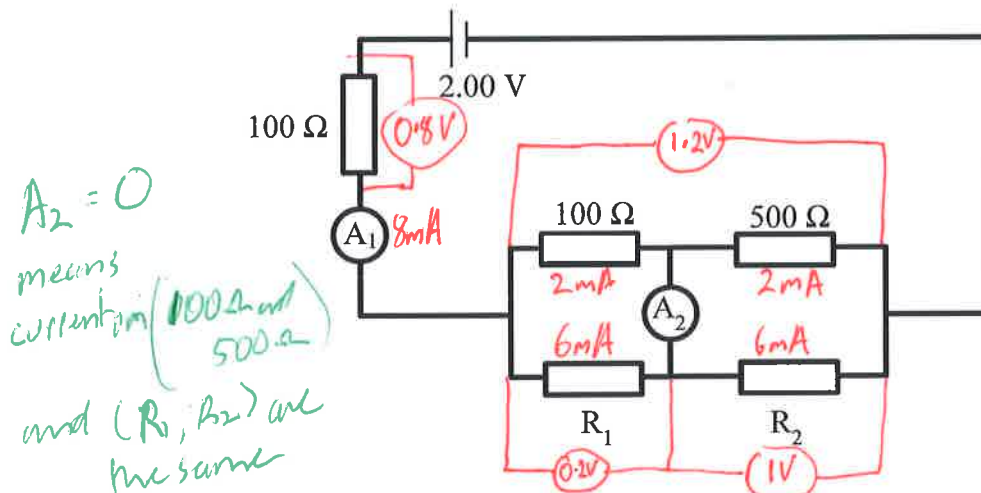
Both forces contain q , so value and sign cancel out.

① Identifies that the net force is zero

(35% of form)
1/1

Question 29 (3 marks)

Consider the following circuit diagram.

Ammeter, A_1 , reads 8.00 mA and ammeter, A_2 reads 0 A.What is the value of R_1 in ohms?

- ① Find total resistance $R_T = \frac{2V}{8mA} = 250 \Omega$. (Resistance of branch = 150Ω) ³
 - ② Find voltage across parallel branch R_p , $V_{100\Omega} = 8mA \times 100 = 0.8V$
 $R_p = 2 \times 0.8 = 1.2V$
 - ③ When $A_2 = 0$ same current in 100Ω & 500Ω ($I = \frac{V}{R} = \frac{1.2}{600} = 2mA$)
 - ④ 8mA in circuit, 2mA in top branch, means 6mA through R_1 and R_2 .
 - ⑤ circuit is still also means same voltage on ($R_1, 100\Omega$) and ($R_2, 500\Omega$)
 voltage on 100Ω resistor $V = IR = 2mA \times 100\Omega = 0.2V$ (same as R_1)
 - ⑥ Resistor R_1 has 6mA current and 0.2V so $R_1 = \frac{V}{I} = \frac{0.2}{6mA} = 33\frac{1}{3} \Omega$
 $(R_2 = \frac{1V}{6mA} = 166\frac{2}{3} \Omega)$
- $R_1 = 33\frac{1}{3} \Omega$ (1 mark)

Marking: There were many paths to the correct solution. Marks were issued for finding relationships that would lead to a solution.

① mark ✓ for any two of each

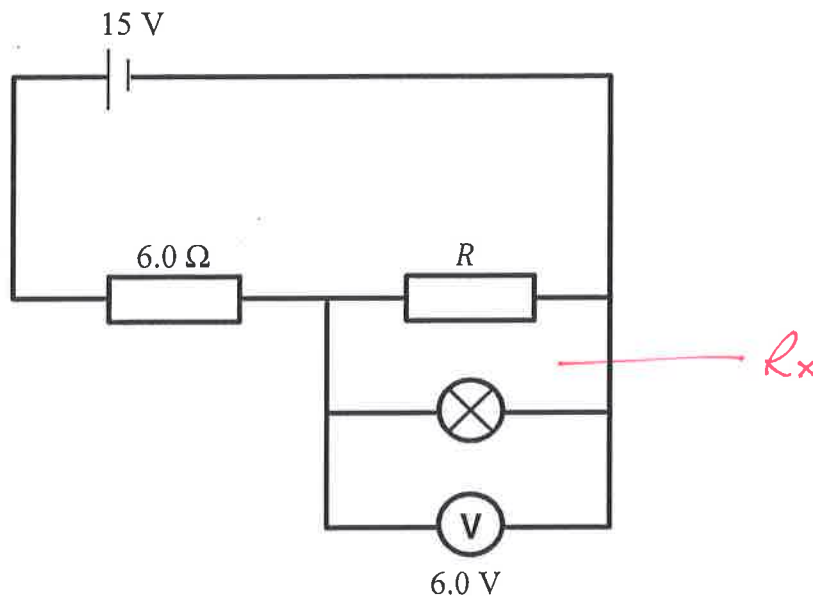
- Resistance of parallel branch is 150Ω
- Voltage of parallel branch is $1.2V$
- $\frac{R_2}{R_1} = 5$

1 mark for $R_1 = 33\frac{1}{3} \Omega$

- Current in each branch is 2mA and 6mA (100Ω & 500Ω) (R_1 and R_2)
- $R_1 + R_2 = 200 \Omega$
- $\frac{R_1}{R_1 + R_2} = \frac{1}{6}$

Question 31 (4 marks)

Consider the following circuit.



A current of 0.60 A flows through the light globe.

- (a) In terms of energy, what is meant by the term '15 V battery'?

1

• ACCEPTED ANYTHING RELATED TO THE DEFINITION OF P.D.
 e.g. $15\text{V} = 15\text{J/C}$ — ①

- (b) Calculate the value of the resistance labelled R.

3

MULTIPLE METHODS:

$R_{\text{TOT}} = 10\Omega$ *	MINUS →	$R_{\text{BULB}} = 6/0.6 = 10\Omega$ *
$R_x = 4\Omega$ *		$I_{\text{TOT}} = \frac{15-6}{6} = 1.5\text{A}$ *

— ①

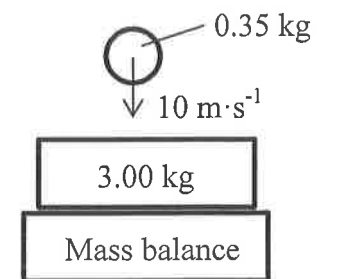
$\therefore I_R = 1.5 - 0.6 = 0.9$ — ②

* ALL POTENTIALLY WORKS ①
 IN ABSENCE OF A CORRECT ANSWER

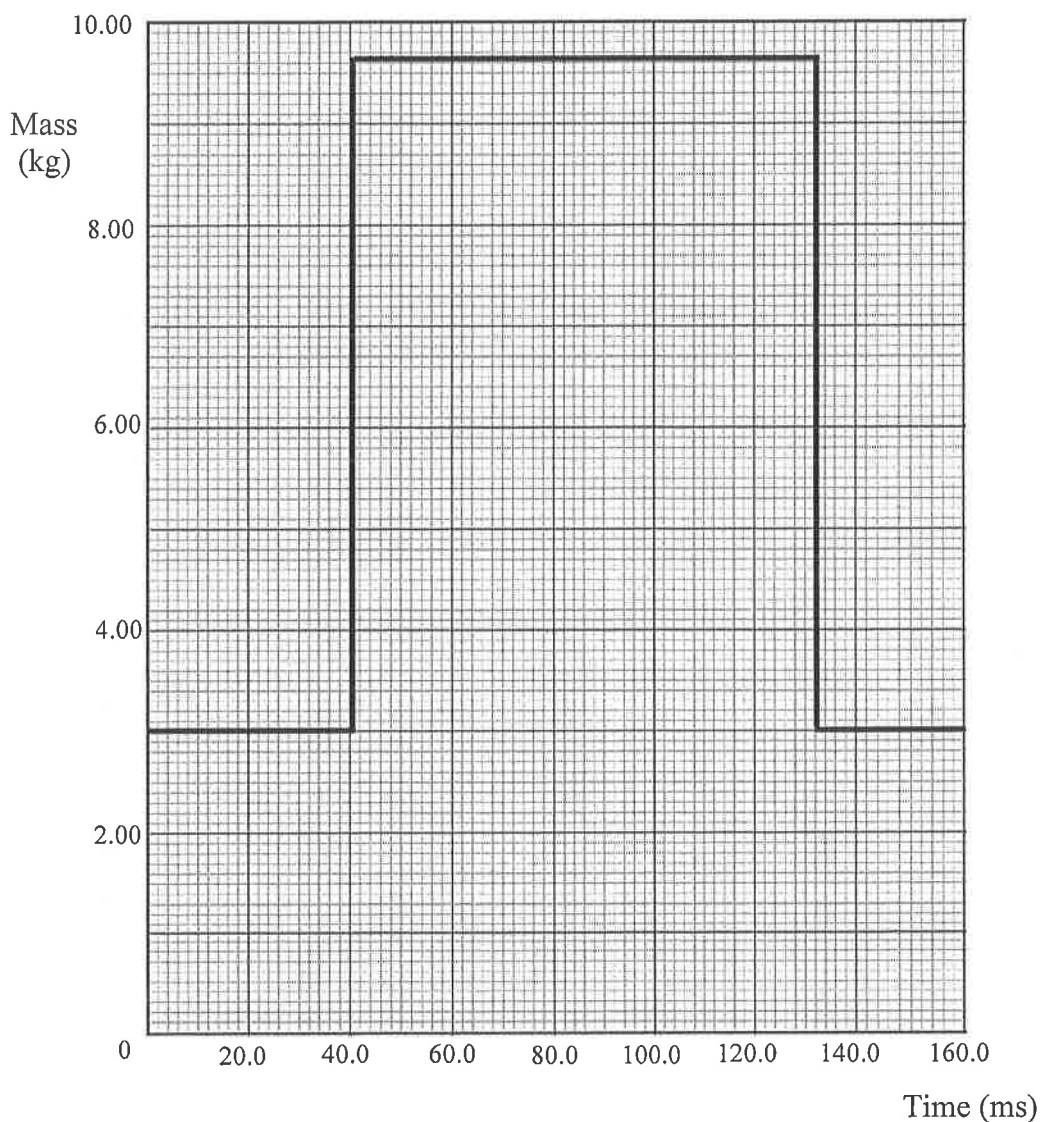
$\therefore R_R = 6/0.9 = 6.67\Omega$ — ③

Question 32 (5 marks)**Marks**

The diagram below shows a ball of mass 0.35 kg falling onto a 3.00 kg block which is sitting on a sensitive mass balance calibrated in kg. The ball hits the 3.00 kg block at a speed of $10\text{ m}\cdot\text{s}^{-1}$.



The graph below shows how the reading on the mass balance varies just before, during and just after the ball hits the block.



Question 32 continued on next page.

Question 32 continued

Marks

- (a) Calculate the magnitude of the momentum of the ball just before it hits the 3.00 kg block.

1

$$= 0.35 \times 10$$

$$= 3.5 \text{ kg ms}^{-1}$$

①

- (b) Calculate how high the ball bounces (relative to the top of the block) after impact with the 3.00 kg block.

4

$$\Delta p = F \times t = (9.6 - 3.00) \times (132 - 40) \times 10^{-3} \times 9.8$$

$$\therefore \Delta p = 5.95 \text{ N s}$$

①

$$\therefore p_{\text{after}} = 5.95 - 3.5 = 2.45 \text{ kg ms}^{-1}$$

②

$$\therefore u_{\text{after}} = \frac{2.45}{0.35} = 7.0 \text{ ms}^{-1}$$

③

$$\therefore v = \frac{v^2 - u^2}{2a} = \frac{7.0^2}{19.6} = 2.5 \text{ m}$$

④

For ① Accepted A
Calculation of v ,
IF BASED on A Reasonable
Assumption (e.g. $u \leq 10 \text{ ms}^{-1}$)

ACCEPT A REASONABLE
Error Margin.

* NB - this should be 3.35, to include the mass of the ball.

NoBODY Included this :: IT WAS NOT PENALISED.

BUT, INCLUDING IT, THE ANSWERS SHOULD BE: $\Delta p = 5.635 \text{ N s}$

$$p_f = 2.135 \text{ kg ms}^{-1}$$

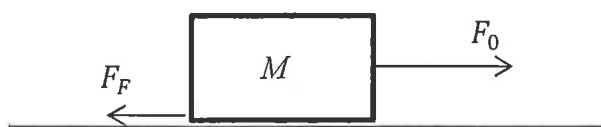
$$u_f = 6.1 \text{ ms}^{-1}$$

$$v = 1.90 \text{ m}$$

actually
I think one boy did,
but he made a mistake elsewhere...

Question 33 (8 marks)**Marks**

A boy performs the following experiment to investigate friction.



The boy pulls blocks of different masses, M , along a horizontal surface with a constant force, F_0 , and measures their acceleration.

The boy assumes that the force of friction, F_F , acting on the blocks is directly proportional to the weight of the blocks, Mg , and is given by the equation:

$$F_F = \mu Mg$$

where μ is called the 'coefficient of friction' and can be assumed to be constant. *Not MARKED, Poor SE*

The results obtained by the boy are given in the table below.

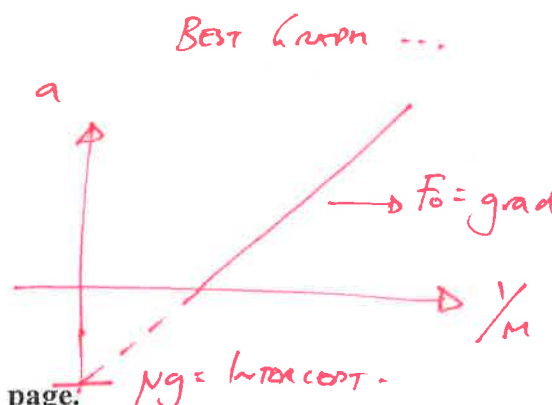
Mass of block (kg)	Acceleration of block ($\text{m}\cdot\text{s}^{-2}$)	$\frac{1}{M}$	
1.00	10.5	1.0	
1.50	6.35	0.67	
2.00	4.39	0.50	
3.00	2.51	0.33	
4.00	1.42	0.25	
5.00	0.78	0.20	

The boy knows that the motion of the blocks is subject to Newton's Second Law, and assumes that the block obeys the relationship:

$$F_0 - \mu Mg = Ma$$

which can be written as

$$a = F_0 \left(\frac{1}{M} \right) - \mu g$$



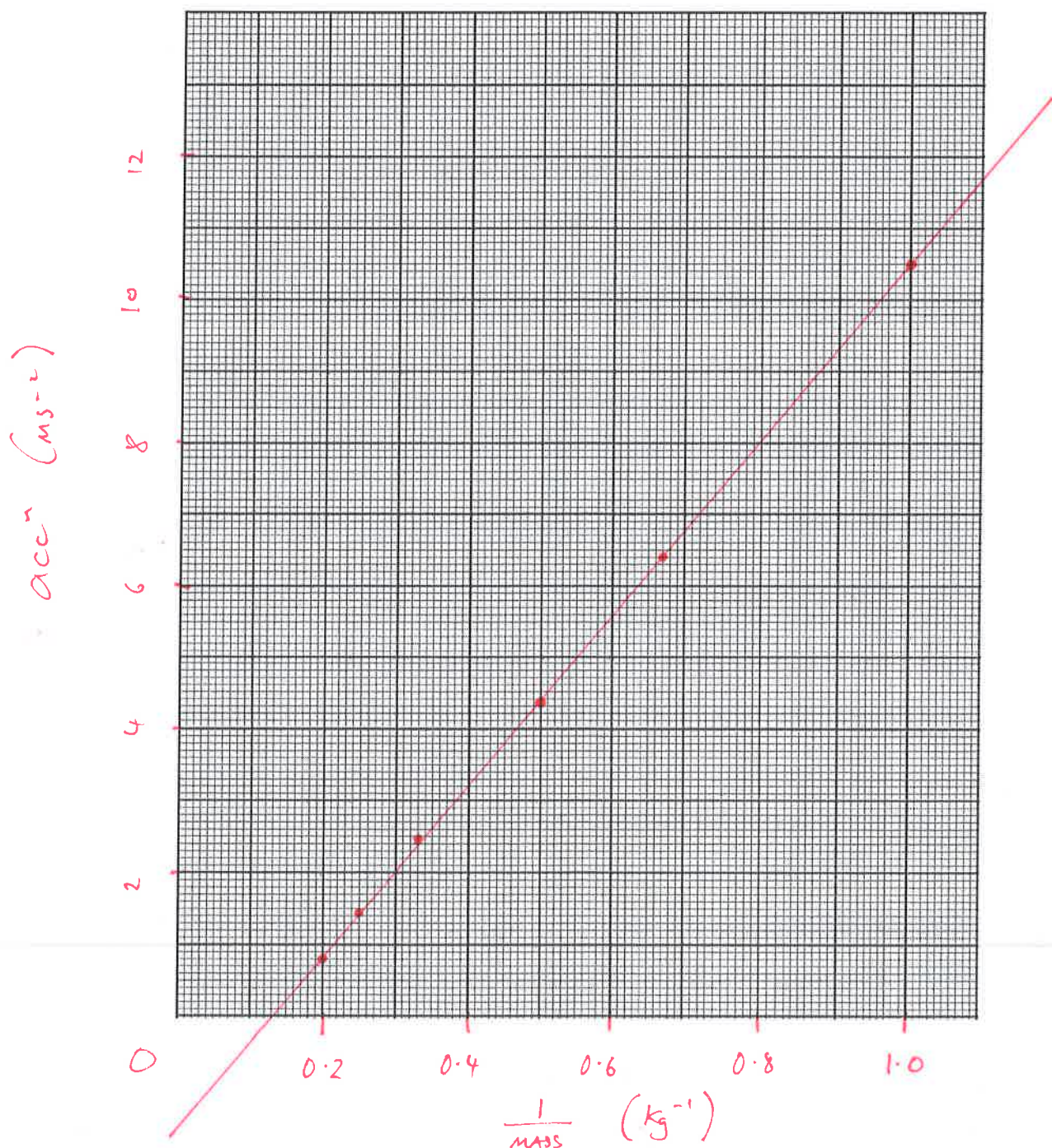
Question 33 continued on next page.

Question 33 continued

Marks

- (a) Plot a **straight line graph** to confirm the relationship between the acceleration and the mass of the blocks, given on the previous page.

(Extra columns have been provided in the table on the previous page, for you to use if you need them.)



4

Question 33 continued on next page.

SEE NEXT PAGE FOR MARKING SCHEME.

Question 33 continued

Marks

(b) Use your graph to determine:

i) the constant force, F_0 .

2

$$F_0 = \text{GRADIENT} = \text{e.g. } \frac{(10.5 - 0)}{(1.0 - 0.13)} = 12.1 \text{ N} \quad \text{--- (2)}$$

(1) /

ACCEPTED REASONABLE RANGE

ii) the coefficient of friction, μ .

2

Either By Intercept or: When $a = 0$ $F_0 \cdot \frac{1}{m} = \mu g$.

$$\frac{1}{m} = 0.13 \quad \therefore F_0 \times 0.13 = \mu \cdot 9.8$$

(1) /

$$\mu = 0.16 \quad \text{--- (2)}$$

a) STANDARD GRAPHING MARK SCHEME: POINTS; AXES; LABELS; LOBF; SCALE; CLARITY
 $\Rightarrow$ (4).

NB1 - IF GRAPH IS NOT A STRAIGHT LINE GRAPH, MAX (2) !!

(e.g. $a \propto m$)

NB2 - NOT ALL STRAIGHT LINE GRAPHS DEMONSTRATE THE "RELATIONSHIP BETWEEN a and m ". !!!

b) IF GRAPH IN a) IS NOT A STRAIGHT LINE THEN b) CAN BE ANSWERED USING SIMULTANEOUS EQUATIONS:

2 x (1) VALID EQUATION

2 x (1) FOR F_0 AND μ .